

Nonlinear Solid Mechanics

A COMPILATION OF
NOTES, NOTATIONS, AND
ILLUSTRATIONS

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Abstract

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Acknowledgements

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Chapter 1

Introduction

1.1 Purpose and Scope

- The purpose of this manuscript is to memorialize the theories of solid dynamics, thermodynamics, and shock physics.
- The scope is simply to include a listing of variables, equations, and methods used. It is not intended to be a substitute for textbooks on continuum mechanics and the finite element method.

1.2 Mathematical Preliminaries

As defined by Gurtin,¹

- LIN= the set of all tensors
- LIN⁺= the set of all sensors $\mathbf{S}$ with $\det \mathbf{S} > 0$
- SYM= set of all symmetric tensors
- SKW= set of all skew tensors
- PSYM= set of all positive definite, symmetric tensors
- ORTH= set of all orthogonal tensors
- ORTH⁺= set of all rotations

As defined by Curnier,² for all second-order tensors $\mathbf{S} \in \text{LIN}$, the **tracor** $\mathbb{T}$ and **deviator** $\mathbb{D}$ are fourth-order tensors are defined as

$$\mathbb{T} \triangleq \frac{1}{3} \delta_{ij} \delta_{kl} \hat{\mathbf{e}}_i \otimes \hat{\mathbf{e}}_j \otimes \hat{\mathbf{e}}_k \otimes \hat{\mathbf{e}}_l \quad (1.1)$$

such that

$$\mathbb{T} : \mathbf{S} = \mathbf{S}_{\text{vol}} \triangleq \frac{1}{3} \text{tr}(\mathbf{S}) \mathbf{1}, \quad (1.2)$$

and

$$\mathbb{D} \triangleq \left(\frac{1}{2} \delta_{ik} \delta_{jl} + \frac{1}{2} \delta_{il} \delta_{jk} - \frac{1}{3} \delta_{ij} \delta_{kl} \right) \hat{\mathbf{e}}_i \otimes \hat{\mathbf{e}}_j \otimes \hat{\mathbf{e}}_k \otimes \hat{\mathbf{e}}_l \quad (1.3)$$

such that

$$\mathbb{D} : \mathbf{S} = \mathbf{S}_{\text{dev}}, \triangleq \mathbf{S} - \frac{1}{3} \text{tr}(\mathbf{S}) \mathbf{1}. \quad (1.4)$$

Note, there are two other fourth-order tensors often used, the **symmetrizer** $\mathbb{S}$ and the **skewer** $\mathbb{W}$ defined, respectively, as

$$\mathbb{S} \triangleq \frac{1}{2} (\delta_{ik} \delta_{jl} + \delta_{il} \delta_{jk}) \hat{\mathbf{e}}_i \otimes \hat{\mathbf{e}}_j \otimes \hat{\mathbf{e}}_k \otimes \hat{\mathbf{e}}_l, \quad (1.5)$$

and

$$\mathbb{W} \triangleq \frac{1}{2} (\delta_{ik} \delta_{jl} - \delta_{il} \delta_{jk}) \hat{\mathbf{e}}_i \otimes \hat{\mathbf{e}}_j \otimes \hat{\mathbf{e}}_k \otimes \hat{\mathbf{e}}_l. \quad (1.6)$$

¹Gurtin ME. An introduction to continuum mechanics. Academic press; 1982, Vol. 158, Jan 12.

²Curnier A. Computational methods in solid mechanics. Springer Science & Business Media; 2012 Dec 6.

Chapter 2

Formulation

2.1 Bodies and Particles

Let body $\mathcal{B}$ be an open, bounded subset of three-dimensional Euclidean space $\mathbb{R}^{n_{\text{sd}}}$ with boundary $\partial\mathcal{B}$. The number of space dimensions, n_{sd} , is assumed equal to three unless stated otherwise. The boundary is decomposed within a particular space dimension into two non-overlapping subdomains, Γ_u and Γ_σ where displacements or tractions are prescribed. Thus $\partial\Omega = \Gamma_u \cup \Gamma_\sigma$ and $\Gamma_u \cap \Gamma_\sigma = \emptyset$. Set closure is denoted $\bar{\mathcal{B}} = \mathcal{B} \cup \partial\mathcal{B}$. Let the body be composed of a closed set of particles with coordinates $\mathbf{X}$ in the reference configuration. The particle has coordinates $\mathbf{X} = \{X_1, X_2, X_3\}^T$ as measured from some origin O in the dextral, orthonormal inertial reference from $\mathbf{E}_1, \mathbf{E}_2, \mathbf{E}_3$.

2.2 Motion

Let the arbitrary time interval be defined as $\mathbb{T} \triangleq [t_0, T]$.¹ Let the motion, a one parameter family of configurations, $\varphi : \mathcal{B} \times \mathbb{T} \mapsto \mathbb{R}^3$ map the material particle $\mathbf{X}$ into the current configuration $\mathbf{x}$ such that

$$\mathbf{x} = \varphi(\mathbf{X}, t) = \varphi_t(\mathbf{X}). \quad (2.1)$$

A motion evaluated at a particular time $t \in \mathbb{T}$ is referred to as a configuration or placement. To each configuration, we assign a displacement field $\mathbf{u} : \mathcal{B} \mapsto \mathbb{R}^3$ such that

$$\mathbf{u}(\mathbf{X}, t) = \varphi(\mathbf{X}, t) - \mathbf{X}. \quad (2.2)$$

Thus, the current configuration φ is simply a function of the original placement $\mathbf{X}$, plus a displacement $\mathbf{u}$, which is **function** of placement $\mathbf{X}$ and time t :

$$\varphi(\mathbf{X}, t) = \mathbf{X} + \mathbf{u}(\mathbf{X}, t). \quad (2.3)$$

It is important to emphasize that the displacement remains a function $\mathbf{u} = \mathbf{u}(\mathbf{X}, t)$ of reference configuration $\mathbf{X}$ and time t . The displacement function $\mathbf{u}$ is not a simple (constant) displacement, which we refer to as “offset” in Section 2.2.2.

¹Note that t_0 , while typically zero, may be any real number.

2.2.1 Linear Motions

In this section, we consider the equation of a line

$$y = f(x) = mx + b, \quad (2.4)$$

as inspiration for writing (2.3) as²

$$\mathbf{x} = \varphi(\mathbf{X}) = \underbrace{\mathbf{M}\mathbf{X}}_{\text{linear in } \mathbf{X}} + \underbrace{\mathbf{b}}_{\text{constant}} \stackrel{\text{set}}{=} \mathbf{X} + \mathbf{u}(\mathbf{X}). \quad (2.5)$$

Then the displacement can be written as

$$\mathbf{u}(\mathbf{X}) = \mathbf{M}\mathbf{X} + \mathbf{b} - \mathbf{X}, \quad (2.6)$$

$$= (\mathbf{M} - \mathbf{1})\mathbf{X} + \mathbf{b}. \quad (2.7)$$

In the following sections, we will be exploring the effect of “slope” $\mathbf{M}$ and “intercept” $\mathbf{b}$ on displacements $\mathbf{u}$. To explore some motions, we shall consider the two-dimensional body shown in Fig. 2.1.

The most elementary displacement is homogeneous, when $\mathbf{M} = \mathbf{1}$ and $\mathbf{b} = \mathbf{0}$, then $\mathbf{u} = \mathbf{0}$. In this case $\varphi = \mathbf{X}$ and the motion is trivial (and uninteresting). Nonetheless, this is a simple first case to consider.

2.2.2 Offset

Next, consider $\mathbf{b} \neq \mathbf{0}$, while retaining $\mathbf{M} = \mathbf{1}$. For the concrete example in Fig. (2.2), let $\mathbf{b} = \langle b_1, b_2 \rangle^\top = \langle 3, 2 \rangle^\top$. In this case, we see the placement φ moves right and up on the page, relative to the reference configuration $\mathbf{X}$, by an amount of 3 and 2, respectively.

²For the time being, we will not be concerned with time t , so we eliminate it from our notation for the sake of brevity.

Figure 2.1: Reference configuration of a body defined by particles with $X_1 \in [-1, 1]$ and $X_2 \in [-1, 1]$. With the continuum body is composed of an infinite number of particles, only nine particles are shown for the sake of simplicity. Eight of the nine particles shown are on the boundary of the body, and a single particle at the geometric center is the ninth particle.

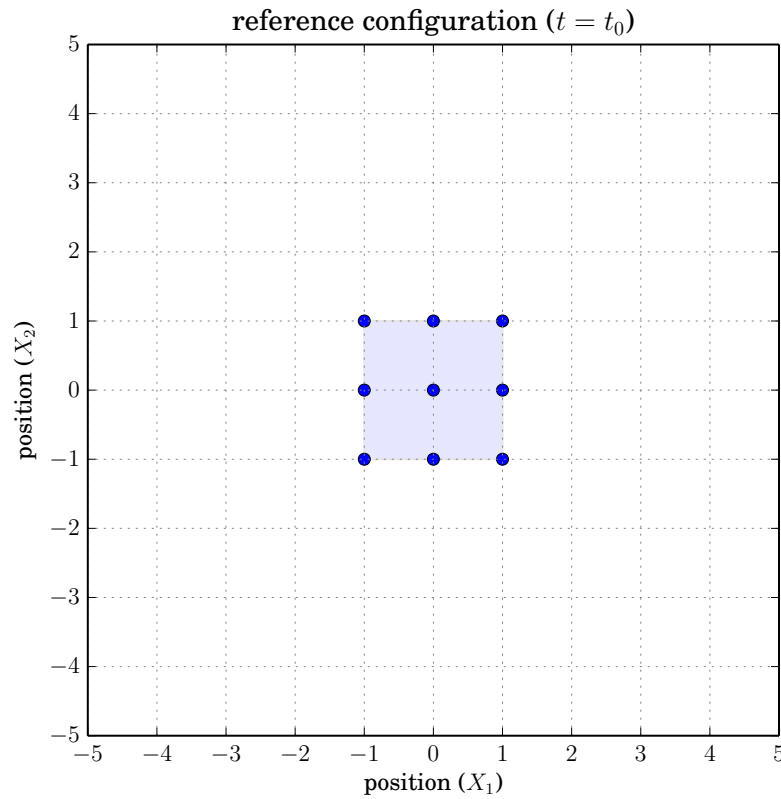
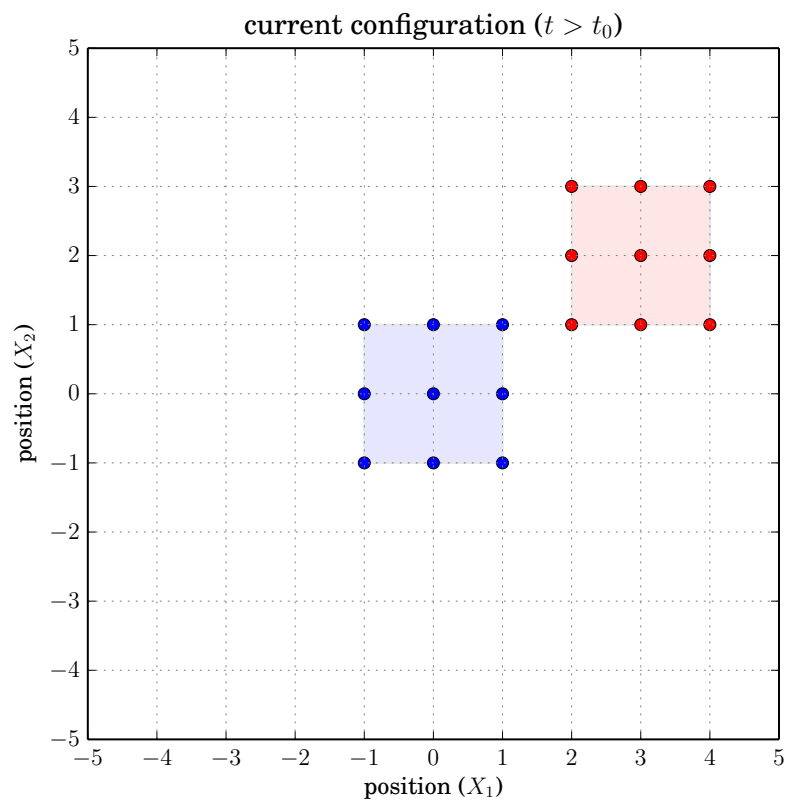


Figure 2.2: Current configuration shown in red, with reference configuration shown in blue and retained for context.



2.2.3 Stretch

Consider now $\mathbf{M} = \lambda \mathbf{1}$ and $\mathbf{b} = \mathbf{0}$. This describes **volumetric** response, also known as a **dilatational** response. The parameter $\lambda \in \mathbb{R}$ causes the current configuration to change volume as

- $\lambda > 0$ expansion,
- $\lambda = 0$ identity,
- $\lambda < 0$ contraction.

Notice that λ describes the “slope” of the mapping from reference to current configurations. In the case that $\lambda \rightarrow +\infty$, the volume grows infinite. In the case that $\lambda \rightarrow -\infty$, the volume grows to zero. For concreteness, consider $\lambda = 2$. The volumetric expansion is shown in Fig. 2.3.

The area in the reference configuration was $2 \times 2 = 4$, whereas the area in the current configuration was $4 \times 4 = 16$.

Next consider a slightly different form of $\mathbf{M}$, such that

$$\mathbf{M} = \begin{bmatrix} \lambda_1 & 0 \\ 0 & \lambda_2 \end{bmatrix}. \quad (2.8)$$

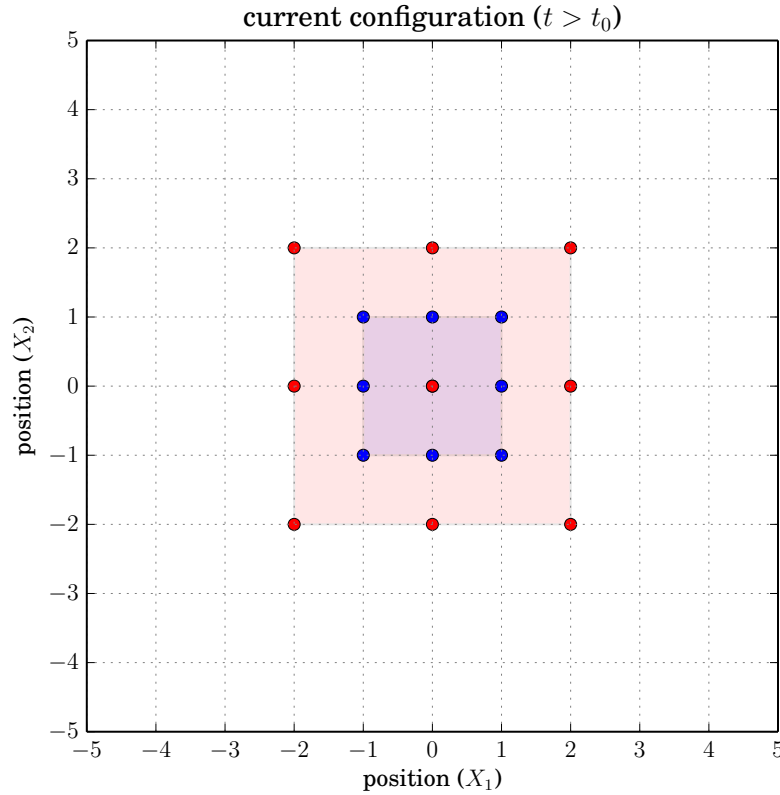
We keep the offset as $\mathbf{b} = \mathbf{0}$.

For concreteness, consider $\lambda_1 = 2$ and $\lambda_2 = \frac{1}{2}$. The combined expansion and compression is shown in Fig. 2.4.

The area in the reference configuration was $2 \times 2 = 4$, whereas the area in the current configuration was $4 \times 1 = 4$. Out of coincidence, we have chosen stretches along the two axes such that the area does not change although the dimensions and thus shape have changed. This is an example of an **isochoric** motion. The roots *iso-* (*equal*) and *choro-* (*space*) describe a process from initial state 0 to final state 1 where the volume remains constant.

Now, the motion as we have described it, doesn’t yet show the *path* taken between the reference and current configuration. To get that detail, we would need to introduce the time parameter t , which we have suppressed for now. However, if we were to say the time interval is simply two points t_0 and t_1 , we can create (the rather course) path between these two states 0 and 1, and connect with the isochoric process *PV* diagram in thermodynamics, shown in Fig. 2.5.

Figure 2.3: Current configuration shown in red, with reference configuration shown in blue and retained for context.



We have yet to introduce and define pressure P , so this connection may be tenuous at the moment. Nonetheless, we introduce this connection now to motivate the single idea that isochoric motions will play an important role in our later work.

2.2.4 Rotation

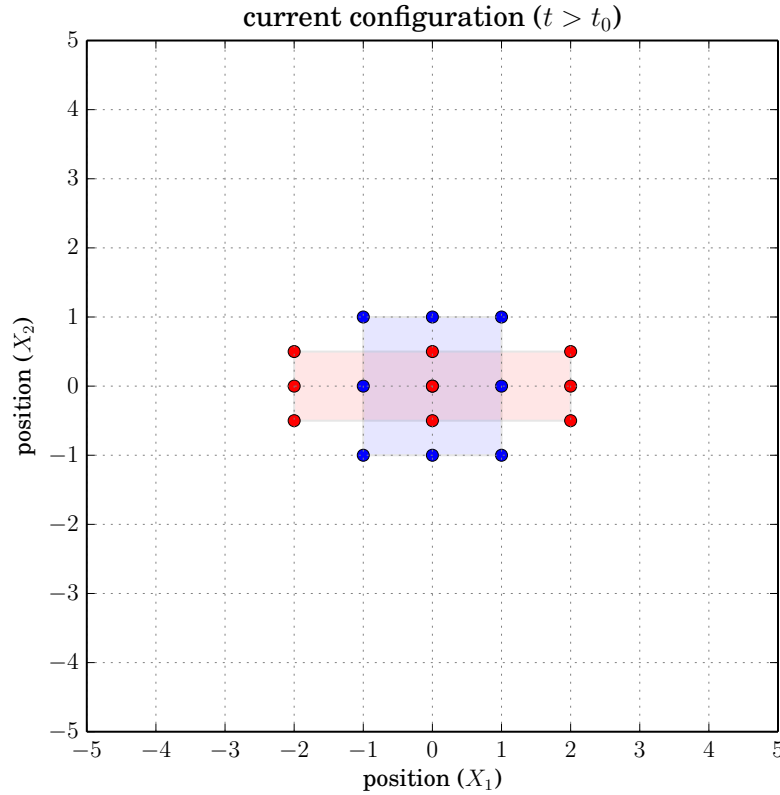
2.3 Deformation Gradient

To each configuration, we define a deformation gradient $\mathbf{F} : \mathcal{B} \times \mathbb{T} \mapsto \mathbb{M}_+^3$ such that

$$\mathbf{F} \triangleq \nabla_0 \varphi \iff F_{iJ} \triangleq \frac{\partial \varphi_i}{\partial X_J}. \quad (2.9)$$

Remarks:

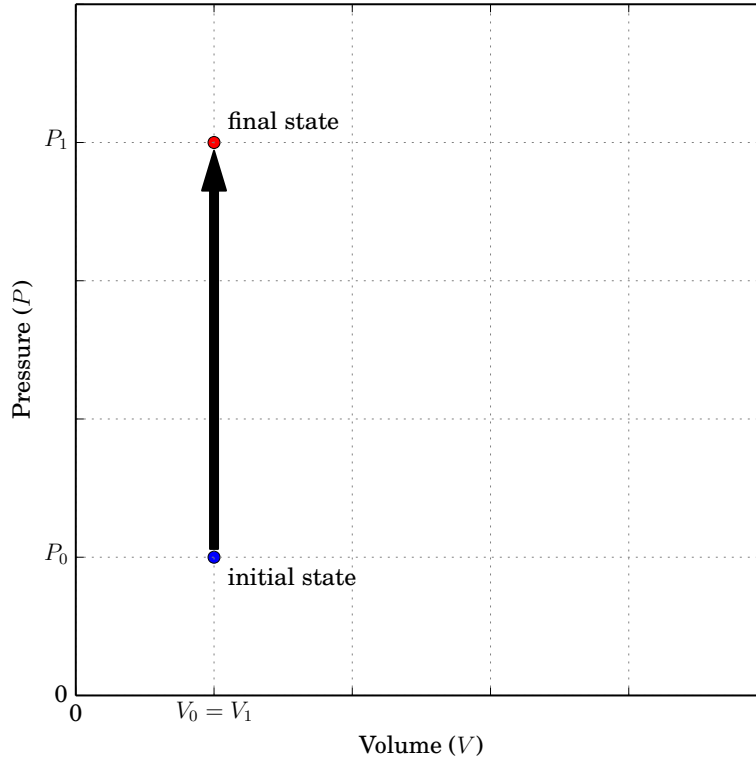
Figure 2.4: Current configuration shown in red, with reference configuration shown in blue and retained for context.



- Lower case indices represent components of tensors in the current configuration whereas upper case indices represent components of tensors in the reference configuration. Real, square matrices of dimension three and positive determinants are denoted $\mathbb{M}_+^3$.
- We define gradients taken in the reference configuration as $\nabla_0(\cdot) = \partial(\cdot)/\partial X_J$ to distinguish from gradients taken in the current configuration, denoted simply as $\nabla(\cdot) = \partial(\cdot)/\partial x_j$.
- All configurations must be admissible in the sense that the Jacobian of the deformation J must be positive

$$J \triangleq \det \mathbf{F} > 0. \quad (2.10)$$

Figure 2.5: Isochoric process from initial state (reference configuration) 0 to final state (current configuration) 1.



This requirement keeps the deformations from mapping the body to a single, infinitesimally small point ($J = 0$) or turning the body inside-out ($J < 0$).

From the displacement field in (2.2), we see that the relationship between the displacement gradient and the deformation gradient is given by

$$\nabla_0 \mathbf{u} = \nabla_0 \boldsymbol{\varphi} - \mathbf{1} = \mathbf{F} - \mathbf{1}. \quad (2.11)$$

2.4 Strain

2.4.1 Right Cauchy-Green

A standard strain measure is the right Cauchy-Green strain tensor

$$\mathbf{C} \triangleq \mathbf{F}^\top \mathbf{F} \iff C_{IJ} \triangleq F_{Ii} F_{iJ}. \quad (2.12)$$

The right Cauchy-Green strain tensor $\mathbf{C}$ gets its name from the fact that the deformation gradient $\mathbf{F}$ sits to the right. Note that this strain tensor is defined in the reference configuration. The right Cauchy-Green strain tensor can be written explicitly as reference gradients of the motion $\boldsymbol{\varphi}$ or displacement $\mathbf{u}$:

$$\mathbf{C} \stackrel{(2.9)}{=} (\nabla_0 \boldsymbol{\varphi})^\top \nabla_0 \boldsymbol{\varphi} \stackrel{(2.11)}{=} (\mathbf{1} + \nabla_0 \mathbf{u})^\top (\mathbf{1} + \nabla_0 \mathbf{u}) \quad (2.13)$$

$$= \mathbf{1} + \nabla_0 \mathbf{u} + (\nabla_0 \mathbf{u})^\top + (\nabla_0 \mathbf{u})^\top \nabla_0 \mathbf{u} \quad (2.14)$$

2.4.2 Left Cauchy-Green

In contrast, a left Cauchy-Green strain tensor $\mathbf{b}$ can be defined with the deformation gradient $\mathbf{F}$ sitting on the left.

$$\mathbf{b} \triangleq \mathbf{F} \mathbf{F}^\top \iff b_{ij} \triangleq F_{iJ} F_{Jj}. \quad (2.15)$$

Note that this strain measure is defined in the current configuration.

2.4.3 Lagrangian (Green) Strain Tensor

The Lagrangian (Green) strain tensor,

$$\mathbf{E} \triangleq \frac{1}{2}(\mathbf{C} - \mathbf{1}) \iff E_{IJ} \triangleq \frac{1}{2}(C_{IJ} - \delta_{IJ}), \quad (2.16)$$

is closely related to the right Cauchy-Green strain tensor and is often used in defining constitutive law relationships because the measure, when linearized about the reference configuration, coincides with the small strain tensor of linear deformation elasticity, denoted $\boldsymbol{\epsilon}$ and defined in (to come). This relationship can be seen as follows:

$$2\mathbf{E} \stackrel{(2.14)}{=} \mathbf{1} + \nabla_0 \mathbf{u} + (\nabla_0 \mathbf{u})^\top + (\nabla_0 \mathbf{u})^\top \nabla_0 \mathbf{u} - \mathbf{1}, \quad (2.17)$$

$$2\mathbf{E}_{\text{LIN}} = \nabla_0 \mathbf{u} + (\nabla_0 \mathbf{u})^\top, \quad (2.18)$$

where the higher-order (quadratic) term in (2.17) is set to zero to achieve (2.18).

2.4.4 Eulerian (Almansi) Strain Tensor

The Eulerian (Almansi) strain tensor,

$$\mathbf{a} \triangleq \frac{1}{2}(\mathbf{1} - \mathbf{b}^{-1}) \iff \mathbf{a}_{ij} \triangleq \frac{1}{2}(\delta_{ij} - (\mathbf{b}_{ij})^{-1}), \quad (2.19)$$

can likewise be used to approximate the small strain tensor $\boldsymbol{\epsilon}$ by combining (2.15) and (2.9) as follows:

$$\begin{aligned} 2\mathbf{a} &= \mathbf{1} - [\nabla_0 \boldsymbol{\varphi} (\nabla_0 \boldsymbol{\varphi})^\top]^{-1} \\ &= \delta_{ij} - \left[\frac{\partial \varphi_i}{\partial X_J} \frac{\partial \varphi_j}{\partial X_J} \right]^{-1} = \delta_{ij} - \left[\frac{\partial X_J}{\partial \varphi_i} \frac{\partial X_J}{\partial \varphi_j} \right] \\ &\stackrel{(2.2)}{=} \mathbf{1} - [(\mathbf{1} - \nabla \mathbf{u})(\mathbf{1} - \nabla \mathbf{u})^\top] \\ &= \nabla \mathbf{u} + \nabla \mathbf{u}^\top - \nabla \mathbf{u} \nabla \mathbf{u}^\top \end{aligned} \quad (2.20)$$

$$2\mathbf{a}_{\text{LIN}} = \nabla \mathbf{u} + (\nabla \mathbf{u})^\top, \quad (2.21)$$

where higher-order (quadratic) term in (2.20) is set to zero to achieve (2.21).

2.4.5 When Strains are Small

2.4.6 Seth-Hill Strain Tensor Family

Seth and Hill showed that the Lagrangian (Green) and Eulerian (Almansi) strain tensors are special cases of the so-called Seth-Hill family of strain measures, defined as

$$\mathbf{E}_{(m)} \triangleq \begin{cases} \frac{1}{m} (\mathbf{U}^m - \mathbf{1}) & \text{for } m \neq 0, \\ \ln \mathbf{U} & \text{for } m = 0. \end{cases} \quad (2.22)$$

The principal stretches $\lambda_k = \ell_k/L_0$, $0 < \lambda_k < \infty$, allow the strain measure $\mathbf{E}_{(m)}$ to be written as principal strains, as a function of principal stretch, $f(\lambda_k)$:

$$\begin{aligned} \mathbf{E}_{(m)} &= \sum_{k=1}^3 f(\lambda_k) \hat{\mathbf{e}}_i \otimes \hat{\mathbf{e}}_i \iff \\ E_{(m)ij} &= f(\lambda_1) \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} + f(\lambda_2) \begin{bmatrix} 0 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix} + f(\lambda_3) \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}, \end{aligned} \quad (2.23)$$

where

$$f(\lambda_k) \triangleq \begin{cases} \frac{1}{m} (\lambda_k^m - 1) & \text{for } m \neq 0, \\ \ln(\lambda_k) & \text{for } m = 0. \end{cases} \quad (2.24)$$

For integer values of $m \in [-2, 2]$, five common strain measures result, listed in Table 2.1, in their three-dimensional and one-dimensional forms.

m	Name	3D	1D
2	Lagrange (Green)	$\mathbf{E}_{(2)} = \frac{1}{2} (\mathbf{U}^2 - \mathbf{1})$	$\mathbf{E}_{\text{LAG}} = \frac{1}{2} \left[\left(\frac{\ell}{L_0} \right)^2 - 1 \right]$
1	Engineering (Biot)	$\mathbf{E}_{(1)} = \mathbf{U} - \mathbf{1}$	$\mathbf{E}_{\text{ENG}} = \frac{\ell - L_0}{L_0}$
0	Log (Hencky, Natural)	$\mathbf{E}_{(0)} = \ln \mathbf{U}$	$\mathbf{E}_{\text{LOG}} = \ln \left(\frac{\ell}{L_0} \right)$
-1	True	$\mathbf{E}_{(-1)} = \mathbf{1} - \mathbf{U}^{-1}$	$\mathbf{E}_{\text{TRUE}} = \frac{\ell - L_0}{\ell}$
-2	Eulerian (Almansi)	$\mathbf{E}_{(-2)} = \frac{1}{2} (\mathbf{1} - \mathbf{U}^{-2})$	$\mathbf{E}_{\text{EUL}} = \frac{1}{2} \left[1 - \left(\frac{L_0}{\ell} \right)^2 \right]$

Table 2.1: Table of strains obtained from the Seth-Hill family.

As will be seen in section (to come),

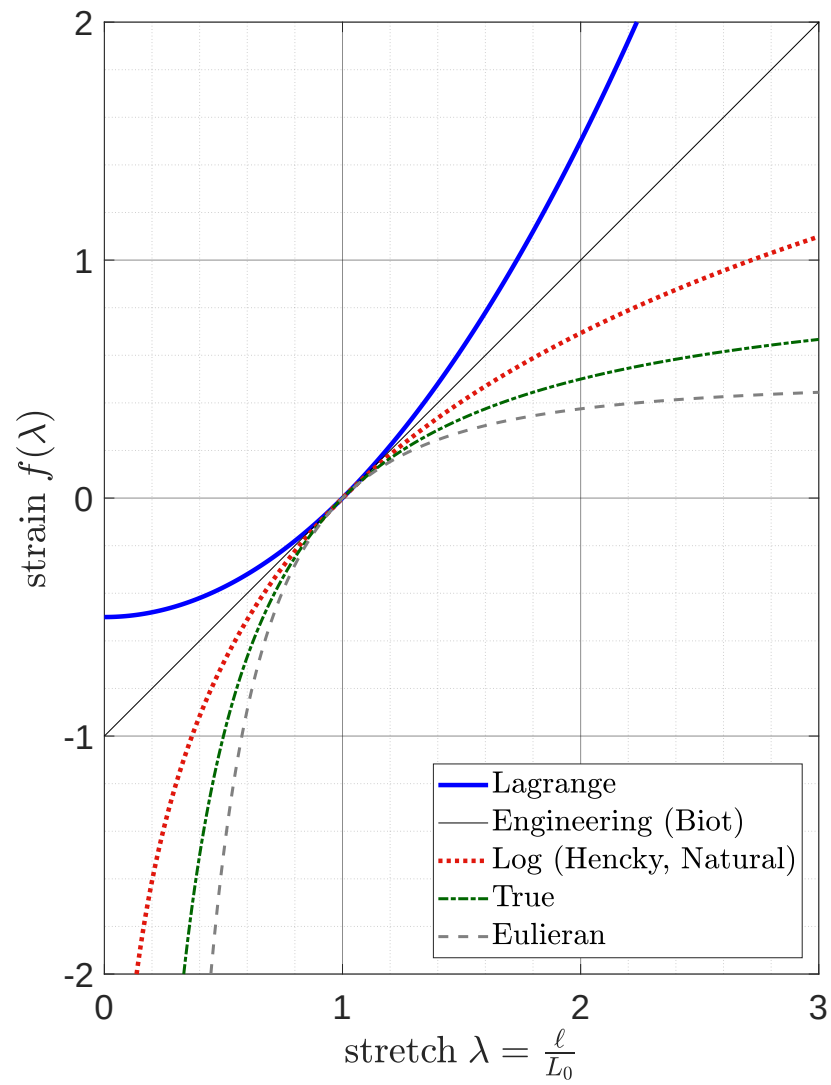
- The Lagrange (Green) strain tensor will be the conjugate of the second Piola-Kirchhoff stress tensor,
- The deformation gradient will be the conjugate of the first Piola-Kirchhoff stress tensor,

- The engineering (Biot) strain tensor will be the conjugate of the Jaumann (Biot) stress tensor,
- The Eulerian (Almansi) strain tensor will be the conjugate of the weighted convected stress tensor.

Figure 2.6 illustrates the one-dimensional strains as a function of stretch ratio $\lambda = \ell/L_0$. The figure illustrates several results:

- For small stretches, $\ell \approx L_0$,
 - The stretch ratio is near unity, $\lambda \approx 1$,
 - The strain values are small, $f(\lambda) \approx 0$,
 - The tangent of the strains with respect to the stretch ratio is near unity, $df/d\lambda \approx 1$.
- For elongations, $\lambda > 1$,
 - The strain monotonically increases since $df/d\lambda > 0$ when $\lambda > 0$.
- For extreme compressions, $\lambda \approx 0$,
 - The Lagrangian (Green) strain goes to a value of $-\frac{1}{2}$,
 - The engineering (Biot) strain tensor goes to a value of -1 ,
 - The Log, true, and Eulerian strains tend to $-\infty$.
- The engineering (Biot) strain is a linear function of stretch λ ; all other measures are nonlinear.

Figure 2.6: One dimensional strain as a function of stretch ratio.



The Linearized Lagrangian (Green) Strain Tensor Creates Nonzero Strains under Rigid Body Rotations

Because it takes on nonzero values under finite rotation, the linearized strain tensor should not be used for geometrically nonlinear analysis. These nonzero values are completely artificial and strictly an artifact of using a linear strain definition with geometrically nonlinear motions. This result is shown as follows:

Let $\mathbf{R} = \mathbf{R}(t)$ be a two-dimensional, rigid body rotation parameterized by time t and scaled by constant $\omega = 2\pi$ radians per second. Then, the motion $\boldsymbol{\varphi}$ of a body can be written as

$$\boldsymbol{\varphi}(\mathbf{X}, t) = \mathbf{R}(t)\mathbf{X} \iff \begin{Bmatrix} x_1 \\ x_2 \end{Bmatrix} = \begin{bmatrix} \cos \omega t & \sin \omega t \\ -\sin \omega t & \cos \omega t \end{bmatrix} \begin{Bmatrix} X_1 \\ X_2 \end{Bmatrix}. \quad (2.25)$$

Then the deformation gradient $\mathbf{F} = \mathbf{F}(t)$ is a function of time alone,

$$\mathbf{F}(t) \iff F_{iJ} = \begin{bmatrix} F_{11} & F_{12} \\ F_{21} & F_{22} \end{bmatrix} = \begin{bmatrix} \cos \omega t & \sin \omega t \\ -\sin \omega t & \cos \omega t \end{bmatrix}. \quad (2.26)$$

The linearized strain tensor is found to be

$$2\mathbf{E}_{\text{LIN}} = \nabla_0 \mathbf{u} + (\nabla_0 \mathbf{u})^T, \quad (2.27)$$

$$= \mathbf{F} - \mathbf{1} + (\mathbf{F} - \mathbf{1})^T, \quad (2.28)$$

$$= F_{iJ} - \delta_{iJ} + F_{Ji} - \delta_{Ji}, \quad (2.29)$$

$$= \begin{bmatrix} \cos \omega t & \sin \omega t \\ -\sin \omega t & \cos \omega t \end{bmatrix} + \begin{bmatrix} \cos \omega t & -\sin \omega t \\ \sin \omega t & \cos \omega t \end{bmatrix} - \begin{bmatrix} 2 & 0 \\ 0 & 2 \end{bmatrix}, \quad (2.30)$$

$$= 2(\cos \omega t - 1)\mathbf{1}. \quad (2.31)$$

Now, for small angles, $\omega t \approx 0$, which is for small deviations $x_1 \approx X_1$, $x_2 \approx X_2$, then $\mathbf{E}_{\text{LIN}} = \mathbf{0}$ for rigid body rotations. However, for arbitrary finite angles, $\omega t \neq 0$, and the linearized strain tensor $\mathbf{E}_{\text{LIN}}$ reports nonzero strain for rigid body rotations, which is nonsensical.

A correct strain tensor will report zero strain for rigid body rotations. One such strain tensor is the fully nonlinear Lagrangian (Green) strain tensor. This result is shown as follows:

$$2\mathbf{E} = \mathbf{F}^T \mathbf{F} - \mathbf{1} \quad (2.32)$$

$$= \begin{bmatrix} \cos \omega t & -\sin \omega t \\ \sin \omega t & \cos \omega t \end{bmatrix} \begin{bmatrix} \cos \omega t & \sin \omega t \\ -\sin \omega t & \cos \omega t \end{bmatrix} - \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = \mathbf{0}. \quad (2.33)$$

2.5 Stress

2.6 Hyperelasticity

2.7 Balance Laws

In this section, we posit the **Master Balance Law** in general form. Also called “conservation of anything,” this description assumes an arbitrary control volume $V \in \mathbb{R}^3$ with boundary ∂V . What follows is the so-called Eulerian formulation, illustrated in Fig. 2.7, which assumes the control volume, while arbitrary in its placement in space, is **fixed** in space.³

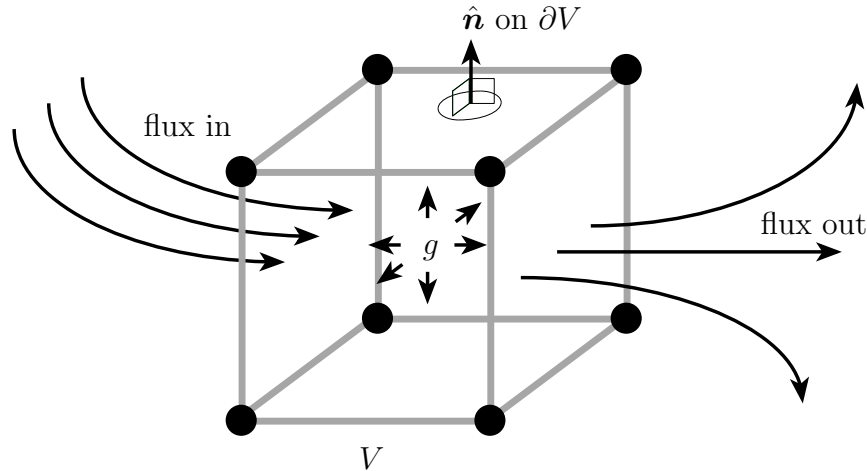


Figure 2.7: Concept of a fixed control volume of the Eulerian formulation.

Now, let

- $\phi(\mathbf{x}, t)$ be the description of some physical quantity per unit volume, which can in general be a scalar, vector, or tensor quantity,
- $\phi^*(\mathbf{x}, t)$ be the generation (depletion) *rate* of the physical quantity per unit volume,
- $\hat{\phi}(\mathbf{x}, t)$ be the physical quantity per unit volume that flows across the boundary with rate (velocity) $\mathbf{v}(\mathbf{x}, t)$, and
- ∂V be the boundary of the control volume, with outward pointing surface normal unit vector $\hat{\mathbf{n}}(\mathbf{x}, t)$.

³The connection to the Lagrangian formulation, in which the control is not fixed in space, but instead, moves through space as the body to which it is attached moves through space, is examined in the sequel.

The time rate of change of the quantity $\phi(\mathbf{x}, t)$ within the differential control volume dV is then equal to

- **generation**—the rate at which the quantity is generated (or depleted) within the control volume by the point source (point sink), plus
- **flux**—the rate at which the quantity is either carried into (or out of) the volume across the volume's differential surface area dA .

These words, stated mathematically, are

$$\underbrace{\frac{d}{dt} \int_V \phi(\mathbf{x}, t) dV}_{\text{rate}} = \underbrace{\int_V \phi^*(\mathbf{x}, t) dV}_{\text{source}} - \underbrace{\int_{\partial V} [\hat{\phi}(\mathbf{x}, t) \mathbf{v}(\mathbf{x}, t)] \cdot \hat{\mathbf{n}} dA}_{\text{flux}}, \quad (2.34)$$

which is the general form of the **Master Balance Law**. We adopt the following sign convention to impart consistency:

- For positive time rate of change of the quantity $\phi(\mathbf{x}, t)$, the generation $\phi^*(\mathbf{x}, t)$ is trivially positive. For quantity $\hat{\phi}(\mathbf{x}, t)$ carried *into* the control volume, the inner product of $\mathbf{v}$ with $\hat{\mathbf{n}}$ is negative, thus a negative outside of the flux integral is used.
- Likewise, for a negative time rate of change of quantity $\phi(\mathbf{x}, t)$, the generation $\phi^*(\mathbf{x}, t)$ is negative. And, for the quantity $\hat{\phi}(\mathbf{x}, t)$ carried *out of* the control volume, the inner product of $\mathbf{v}$ with $\hat{\mathbf{n}}$ is positive, thus the negative outside of the flux integral provides sign consistency.

2.7.1 Balance Law Metaphor

Here we present the Master Balance Law by way of metaphor. Let the volume V be the geographic region of Oregon, with boundary ∂V that touches the state borders of Washington, Idaho, Nevada, and California in the United States.⁴ Let ϕ be the number of people living in the state of Oregon. The change in population $\Delta\phi$ over some time interval Δt is equal to the net generation rate ϕ^* (number of births less the number of deaths in the time interval), plus the net population flux rate $\hat{\phi}\mathbf{v}$ (the number of people moving to Oregon less the number of people moving from Oregon within the time interval).

⁴In this oversimplified metaphor, we assume flux does not occur via the western seaboard nor via the air.

2.7.2 Conservation of Mass

Let us now apply the Master Balance Law to the physical quantity of mass, and develop a relationship for conservation of mass, a concept where mass is neither created nor destroyed. When we speak of the physical quantity of mass per unit volume, we speak of density $\rho(\mathbf{x}, t)$. Thus, we substitute $\rho(\mathbf{x}, t)$ for $\phi(\mathbf{x}, t)$ in the foregoing Master Balance Law.

For mass to be conserved, mass within a control volume is neither created nor destroyed, thus the source term is identically zero. Thus, increases (decreases) in the mass rate term are produced only by flux of mass in through (out from) the boundary. The idea, stated mathematically, is

$$\underbrace{\frac{d}{dt} \int_V \rho(\mathbf{x}, t) dV}_{\text{rate}} = \underbrace{\int_V \rho^*(\mathbf{x}, t) dV}_{\text{source} = 0} - \underbrace{\int_{\partial V} [\rho(\mathbf{x}, t) \mathbf{v}(\mathbf{x}, t)] \cdot \hat{\mathbf{n}}(\mathbf{x}, t) dA}_{\text{flux}}. \quad (2.35)$$

The Divergence Theorem is used to write the flux term surface integral as a volume integral, thus

$$\underbrace{\frac{d}{dt} \int_V \rho(\mathbf{x}, t) dV}_{\text{rate}} = 0 - \underbrace{\int_V \nabla \cdot [\rho(\mathbf{x}, t) \mathbf{v}(\mathbf{x}, t)] dV}_{\text{flux}}. \quad (2.36)$$

Because V is fixed in an inertial frame, the foregoing equation can be rewritten with the rate time derivative inside the integral, thus

$$\int_V \left(\frac{\partial \rho(\mathbf{x}, t)}{\partial t} + \nabla \cdot [\rho(\mathbf{x}, t) \mathbf{v}(\mathbf{x}, t)] \right) dV = 0. \quad (2.37)$$

Finally, arbitrariness of B_t requires that the integrand is identically zero, thus

$$\frac{\partial \rho(\mathbf{x}, t)}{\partial t} + \nabla \cdot [\rho(\mathbf{x}, t) \mathbf{v}(\mathbf{x}, t)] = 0 \iff \rho_{,t} + (\rho v_i)_{,i} = 0, \quad (2.38)$$

which is the **continuity equation**. Some texts will recast the continuity equation by defining a **material derivative** of density as follows: Since $\rho = \rho(\mathbf{x}, t)$, then

$$\dot{\rho} = \frac{\partial \rho}{\partial \mathbf{x}} \frac{\partial \mathbf{x}}{\partial t} + \frac{\partial \rho}{\partial t} \triangleq \frac{D\rho}{Dt}. \quad (2.39)$$

Expansion of (2.38) then gives

$$\frac{\partial \rho}{\partial t} + \nabla \rho \cdot \mathbf{v} + \rho \nabla \cdot \mathbf{v} = \underbrace{\frac{\partial \rho}{\partial t} + \frac{\partial \rho}{\partial \mathbf{x}} \frac{\partial \mathbf{x}}{\partial t}}_{\triangleq \dot{\rho}} + \rho \nabla \cdot \mathbf{v} = 0, \quad (2.40)$$

or simply

$$\dot{\rho} + \rho \nabla \cdot \mathbf{v} = 0. \quad (2.41)$$

2.7.3 Conservation of Linear Momentum**2.7.4 Conservation of Angular Momentum****2.7.5 Conservation of Energy**

2.8 Elastodynamics

2.8.1 Nonlinear Elastodynamics

The bulk modulus B , a measure of local compressibility of a substance, is defined as the ratio of infinitesimal change in pressure dP caused by an infinitesimal change in volume dV , at a given volume V

$$B \triangleq -V \frac{dP}{dV}. \quad (2.42)$$

We require $B > 0$, and since $V > 0$, then $dP/dV < 0$. This results in behaviors observed physically, namely that, all else being equal,

- increase in pressure causes decrease in volume, and vice versa, $\uparrow P \iff \downarrow V$, and
- increase in volume causes decrease in pressure, and vice versa, $\uparrow V \iff \downarrow P$.

Using the definition of density $\rho \triangleq m/V$,

$$\frac{d\rho}{dV} = \frac{-m}{V^2}, \quad (2.43)$$

and

$$\frac{dP}{dV} = \frac{dP}{d\rho} \frac{d\rho}{dV}, \quad (2.44)$$

the bulk modulus B may also be written as

$$B = \rho \frac{dP}{d\rho}. \quad (2.45)$$

Since $B > 0$ and $\rho > 0$, then $dP/d\rho > 0$. This result physically means that, all else being equal,

- increases in pressure cause increases in density (and vice versa), and
- increases in density cause increases in pressure (and vice versa).

2.8.2 Linear Elastodynamics

Linear elastodynamics begins with nonlinear elastodynamics, and adds progressively simplifying (and thus, restrictive) assumptions. The first assumption will be regarding linearity of displacement gradients.

• **Assumption 1:** Second-order and higher displacement gradients are negligible, which restricts the stress-response function to be a special case of the fully general stress response function

$$\mathbf{P} = \mathbf{P}(\mathbf{F}). \quad (2.46)$$

Rewriting the foregoing stress-response function in terms of the displacement gradient,

$$\mathbf{P}(\mathbf{F}) = \mathbf{P}(\mathbf{1} + \nabla_0 \mathbf{u}), \quad (2.47)$$

gives an expression that can be cast in terms of a Taylor series expansion,

$$f(x_0 + \Delta x) = f(x_0) + f'(x_0)\Delta x + \frac{f''(x_0)}{2!}(\Delta x)^2 + \text{H.O.T.}, \quad (2.48)$$

$$= \sum_{j=0}^{\infty} \frac{f^{(j)}(x_0)}{j!} (\Delta x)^j. \quad (2.49)$$

Then

$$\mathbf{P}(\mathbf{1} + \nabla_0 \mathbf{u}) = \mathbf{P}(\mathbf{1}) + \frac{\partial \mathbf{P}(\mathbf{1})}{\partial \mathbf{F}} \nabla_0 \mathbf{u} + \mathcal{O}[(\nabla_0 \mathbf{u})^2]. \quad (2.50)$$

Next, we make another assumption,

• **Assumption 2:** The initial state is stress free, thus $\mathbf{P}(\mathbf{1}) = \mathbf{0}$.

The fourth-order elasticity tensor $\mathbb{C}$ is defined as,

$$\mathbb{C} \triangleq \frac{\partial \mathbf{P}(\mathbf{1})}{\partial \mathbf{F}}. \quad (2.51)$$

Symmetry of the elasticity tensor provides

$$\mathbb{C} : \nabla_0 \mathbf{u} = \mathbb{C} : \text{SYM}(\nabla_0 \mathbf{u}). \quad (2.52)$$

Finally,

$$\text{SYM}(\nabla_0 \mathbf{u}) = \frac{1}{2}(\nabla_0 \mathbf{u} + \nabla_0 \mathbf{u}^\top) = \boldsymbol{\epsilon}. \quad (2.53)$$

These results give

$$\mathbf{P}(\mathbf{1} + \nabla_0 \mathbf{u}) = \mathbf{0} + \mathbb{C} : \boldsymbol{\epsilon} + \text{H.O.T.}, \quad (2.54)$$

or simply

$$\mathbf{P}(\mathbf{1} + \nabla_0 \mathbf{u}) \stackrel{(\text{LIN})}{=} \mathbf{P}_{\text{LIN}}(\boldsymbol{\epsilon}) = \mathbb{C} : \boldsymbol{\epsilon}. \quad (2.55)$$

Next, we can make another limiting assumption,

- **Assumption 3:** The material is **isotropic**,

which allows the stress-reponse function to be written in terms of two Lamé parameters λ and μ as

$$\mathbf{P}_{\text{LIN}}(\boldsymbol{\epsilon}) = \lambda \operatorname{tr}(\boldsymbol{\epsilon}) \mathbf{1} + 2 \mu \boldsymbol{\epsilon}. \quad (2.56)$$

Next, we make another assumption,

- **Assumption 4:** The material is **homogeneous**,

thus ρ_0 , λ , and μ are constants with respect to the position $\mathbf{X}$ in the body.

In the conservation of momentum equation, the divergence of the stress tensor appears. Thus, we find

$$\nabla_0 \cdot \mathbf{P}_{\text{LIN}}(\boldsymbol{\epsilon}) = \nabla_0 \cdot (\lambda \operatorname{tr}(\boldsymbol{\epsilon}) \mathbf{1} + 2 \mu \boldsymbol{\epsilon}) \quad (2.57)$$

$$= \lambda \nabla_0 (\nabla_0 \cdot \mathbf{u}) + \mu [\nabla_0 \cdot (\nabla_0 \mathbf{u}) + \nabla_0 (\nabla_0 \cdot \mathbf{u})], \quad (2.58)$$

$$= (\lambda + \mu) \nabla_0 (\nabla_0 \cdot \mathbf{u}) + \mu \nabla_0 \cdot (\nabla_0 \mathbf{u}). \quad (2.59)$$

Now, since

$$\nabla_0 \cdot (\nabla_0 \mathbf{u}) = \nabla_0 (\nabla_0 \cdot \mathbf{u}) - (\nabla_0 \times \nabla_0 \times \mathbf{u}), \quad (2.60)$$

we can write

$$\nabla_0 \cdot \mathbf{P}_{\text{LIN}}(\boldsymbol{\epsilon}) = \underbrace{(\lambda + 2\mu) \nabla_0 (\nabla_0 \cdot \mathbf{u})}_{\text{dilatational}} - \underbrace{\mu (\nabla_0 \times \nabla_0 \times \mathbf{u})}_{\text{rotational}} \quad (2.61)$$

Finally, we make another assumption

- **Assumption 5:** The **body force** is **negligible**, thus $\mathbf{b}_0 = \mathbf{0}$.

This final assumption allows the **progressive wave equation for solids** to be written as

$$\rho_0 \ddot{\mathbf{u}} = \underbrace{(\lambda + 2\mu) \nabla_0 (\nabla_0 \cdot \mathbf{u})}_{\text{dilatational}} - \underbrace{\mu (\nabla_0 \times \nabla_0 \times \mathbf{u})}_{\text{rotational}} \quad (2.62)$$

There are two wave speeds: the longitudinal sound speed c_L and the transverse sound speed c_T .

$$c_L = \sqrt{\frac{\lambda + 2\mu}{\rho_0}} \quad (\text{longitudinal or P-wave}), \quad (2.63)$$

$$c_T = \sqrt{\frac{\mu}{\rho_0}} \quad (\text{transverse or S-wave}). \quad (2.64)$$

The longitudinal sound speed is faster than the transverse sound speed.

The bulk modulus B is related to the Lamé constants λ and μ through

$$B = \lambda + \frac{2\mu}{3}. \quad (2.65)$$

Substituting for B in the longitudinal sound speed gives

$$c_L = \sqrt{\frac{B + 4\mu/3}{\rho_0}} \quad (\text{longitudinal or P-wave}). \quad (2.66)$$

2.9 Plasticity

We first focus on plasticity in one dimension, as opposed to fully three-dimensional plasticity, to emphasize concepts while being as mathematically simple as possible. After the concepts are clear, the extension of the mathematics from one- to three-dimensions can be made naturally.

2.9.1 Perfect Plasticity

The extension of elasticity to plasticity is made by way of *perfect* plasticity. Conceptually, perfect plasticity in the constitutive stress-strain relationship is obtained by “capping” the elastic response function at some yield stress (to be defined later as σ_y). Beyond the yield stress, the stress response function is perfectly horizontal. Increases in strain produce no increase in stress beyond the yield stress.

Deformation Measures

Let the body Ω have strain measure $\varepsilon : \Omega \rightarrow \mathbb{R}$ in one dimension be defined as a ratio of change in length, that is the difference between the new (deformed, current configuration) length ℓ of body Ω and the original (undeformed, reference configuration) length L , divided by the original length,

$$\varepsilon \triangleq \frac{\ell - L}{L}. \quad (2.67)$$

The total strain ε in the body be defined as an additive decomposition of elastic strain ε^e and plastic strain ε^p ,

$$\varepsilon \triangleq \varepsilon^e + \varepsilon^p. \quad (2.68)$$

Constitutive Law

Then, the traditional stress-strain relationship, $\sigma = E\varepsilon^e$, can be rewritten as

$$\sigma = E(\varepsilon - \varepsilon^p). \quad (2.69)$$

A change in the configuration of the plastic element (see Fig. 2.8) can only occur if the notion of **slip rate** is allowed.

That is, *assume* that $\varepsilon^p(t)$ is a function of time on interval $[0, T] \subset \mathbb{R}$. Then from (2.68) and (2.69), time evolution expressions $\varepsilon(t)$ and $\sigma(t)$ exist, respectively.

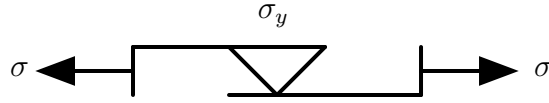


Figure 2.8: Phenomenological model of a friction element. A unit cross-sectional area is assumed. Motion in the friction element occurs only if the friction force, σ_y , is exceeded. The motion, in rate form, is referred to as the slip rate. Note that we employ a slight abuse of notation by using σ to describe a force F , since the cross-sectional area A is unity, $F = \sigma A \implies \sigma \cdot 1 = \sigma$.

Finally, the presence of evolution equations allows the constitutive law in (2.69) to be written as

$$\dot{\sigma} = E (\dot{\varepsilon} - \dot{\varepsilon}^p). \quad (2.70)$$

This form of the constitutive equation will become useful for quantifying plastic flow, as seen below in (2.79).

Yield Function

The stress σ in the frictional device can never greater in absolute value than the yield stress (also known as flow stress) $\sigma_y \in \mathbb{R}^+$. The **yield function** Φ is then defined as

$$\Phi(\sigma) \triangleq |\sigma| - \sigma_y \leq 0. \quad (2.71)$$

For stress levels within the elastic domain, only elastic straining can occur, thus

$$\text{If } \Phi(\sigma) < 0 \Rightarrow \dot{\varepsilon}^p = 0. \quad (2.72)$$

For stress levels on the boundary of the elastic domain (at the yield stress), either plastic strain via plastic loading or non-plastic strain via elastic unloading occurs.

$$\begin{aligned} \text{If } \Phi(\sigma) = 0 \\ \Rightarrow \begin{cases} \dot{\varepsilon}^p \neq 0 & \text{for plastic loading,} \\ \dot{\varepsilon}^p = 0 & \text{for elastic unloading.} \end{cases} \end{aligned} \quad (2.73)$$

Plastic Flow Rule, Loading/Unloading Conditions

We next define how, on the yield surface, the plastic strain evolves through time, which is called the **plastic flow rule**. A reasonable first approximation for the plastic strain evolution is to posit

$$\dot{\epsilon}^p \triangleq \dot{\gamma} \text{sign}(\sigma), \quad (2.74)$$

where $\dot{\gamma} \geq 0$ is the **plastic multiplier** or absolute value of the plastic slip rate, and where $\text{sign} : \mathbb{R} \rightarrow \{-1, +1\}$ is the *signum* function defined as

$$\text{sign}(a) = \begin{cases} +1 & \text{if } a \geq 0, \\ -1 & \text{if } a < 0. \end{cases} \quad (2.75)$$

Both σ and $\dot{\gamma}$ are restricted by **unilateral constraints**,

$$\begin{aligned} \Phi(\sigma) &\leq 0 \\ \dot{\gamma} &\geq 0 \end{aligned} \quad (2.76)$$

Now if $\Phi(\sigma) < 0$, then $\dot{\gamma} = 0$. Similarly, if $\dot{\gamma} > 0$, then $\Phi(\sigma) = 0$. These two observations require

$$\dot{\gamma} \Phi(\sigma) = 0, \quad (2.77)$$

which is referred to as the **complementarity condition**, and when taken with the unilateral constraints, describes the so-called **Kuhn-Tucker conditions**. The unilateral constraints and the complementarity condition are illustrated in Fig. 2.9.

Together, the unilateral constraints and complementarity condition create the normal plasticity law, shown in Fig. 2.10.

Next, consider any time interval $[t, t + \Delta t]$ during which *plastic flow occurs*, the yield function remains constant, $\Phi(\sigma) = 0$. Similarly, if during this time interval, we have elastic response only, and $\dot{\Phi}(\sigma) < 0$, then necessarily $\dot{\gamma} = 0$. These relationships create the **persistency (or consistency) condition**

$$\left. \begin{aligned} \text{If } \dot{\gamma} > 0 &\Rightarrow \dot{\Phi} = 0, \\ \text{If } \dot{\Phi} < 0 &\Rightarrow \dot{\gamma} = 0, \end{aligned} \right\} \Rightarrow \dot{\gamma} \dot{\Phi} = 0. \quad (2.78)$$

Determination of Plastic Multiplier

As of yet, the value of the plastic multiplier $\dot{\gamma}$ has remained undetermined. For non-trivial $\dot{\gamma}$, we know $\dot{\Phi} = 0$ by the persistency condition. An explicit form for $\dot{\Phi}$ may

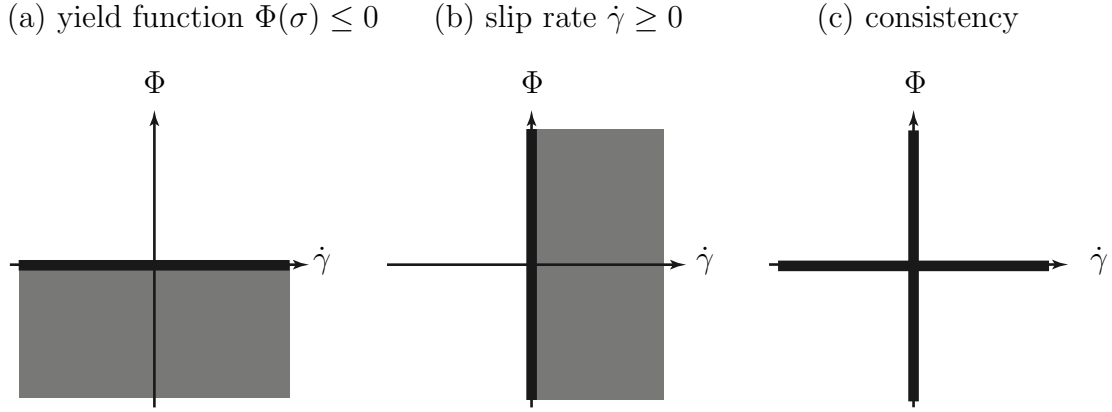


Figure 2.9: Unilateral conditions (a) $\Phi(\sigma) \leq 0$ and (b) slip rate $\dot{\gamma} \geq 0$. The complementarity condition, $\dot{\gamma} \Phi(\sigma) = 0$, is shown in (c).

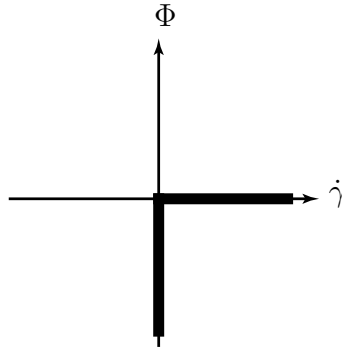


Figure 2.10: The normal plastic law described as the intersection of the unilateral constraints and the complementarity condition.

be found using the chain rule with (2.70) and (2.74),

$$\begin{aligned}
 \frac{\partial \Phi}{\partial t} &= \frac{\partial \Phi}{\partial \sigma} \frac{\partial \sigma}{\partial t} \\
 &= \frac{\partial}{\partial \sigma} (|\sigma| - \sigma_y) E (\dot{\varepsilon} - \dot{\varepsilon}^P) \\
 &= \text{sign}(\sigma) E (\dot{\varepsilon} - \dot{\gamma} \text{sign}(\sigma)),
 \end{aligned} \tag{2.79}$$

or, compactly

$$\dot{\Phi} = \text{sign}(\sigma) E \dot{\varepsilon} - E \dot{\gamma} \stackrel{\text{set}}{=} 0, \tag{2.80}$$

gives

$$\dot{\gamma} = \dot{\varepsilon} \operatorname{sign}(\sigma) \quad (2.81)$$

which, when compared with (2.74), shows that for perfect plasticity during plastic flow,

$$\varepsilon = \varepsilon^P. \quad (2.82)$$

The results from (2.81) and (2.82) are remarkable for the following reasons: During plastic flow,

- None of the strain ε is *elastic* strain ε^e ; 100% of the strain ε is *plastic* strain ε^P . This feature will not be true, in general, for more sophisticated yield functions (for example, that have plastic hardening).
- The plastic flow parameter $\dot{\gamma}$ is equal to the strain ε times the sign of the stress. For more sophisticated yield functions, the plastic flow parameter will not be identically equal to the signed strain, but will be related thereto via factors of the elastic and hardening parameters, as shown subsequent developments and in particular in (2.89).

2.9.2 Hardening Plasticity

Hardening plasticity is considered an extension of perfect plasticity. With perfect plasticity, the slope of stress-strain relationship at yielding is identically zero. With hardening plasticity, the slope becomes positive.⁵

Remarks on Loss of Stability

Perfect plasticity allows for the notion of unbounded strains when stress is at the yield stress. The slope of the stress-strain curve for strains beyond the elastic strain region is identically zero. This zero slope causes numerical problems for the consistent tangent, which is inverted and multiplied by the external load to obtain displacements. Three strategies to circumvent this zero tangent are mentioned here: arc-length, displacement control, and dynamic formulation.

The most notable of the continuation methods is called the **arc-length method**, which is a non-trivial area of study within computational mechanics. Arc-length

⁵For completeness, we note that softening plasticity occurs when the slope becomes negative.

methods will be used eventually, but their explication is delayed for now, in favor of development of hardening plasticity.

Despite being slightly more complex than perfect plasticity, hardening plasticity does not likewise suffer from loss of ellipticity and therefore is actually easier to solve since algorithmic enhancements, such as arc-length, are (usually⁶) unnecessary.

Next, algorithmic formulations that utilize **displacement control** instead of force control can avoid loss of ellipticity if the displacement boundary conditions reduce the number of equations in the tangent to provide rank sufficiency. However, displacement control is less-favored than force control since we often know imposed forces and seek to discover the displacement response. The setting of known inhomogeneous displacements and unknown forces is less frequently encountered in practice. On the topic of load control versus displacement control, de Borst *et al.*,⁷ offer similar insights and additional context.

Finally, elastoplastic **dynamics**, and so-called dynamic relaxation as a subset thereof, are additional strategies to circumvent the above-mentioned problem as these approaches condition the consistent tangent with mass matrices. The benefit of mass matrices comes at the cost of increased expense to compute and store the same, and to incorporate a time stepping algorithm, such as the Newmark methods.

Yield Function

The departure of hardening plasticity from perfect plasticity begins with the yield function. The **yield function** from perfect plasticity in (2.71) is now modified to include a **linear hardening modulus** $H \in \mathbb{R}$ and the **internal hardening variable** $\alpha : [0, T] \rightarrow \mathbb{R}$,

$$\Phi(\sigma, \varepsilon^p) \triangleq |\sigma| - (\sigma_y + H\alpha) \leq 0. \quad (2.83)$$

for strain-hardening plasticity. The variable $\alpha : [0, T] \rightarrow \mathbb{R}$ is called the **internal hardening variable** and is a nonnegative function of the amount of plastic flow

⁶If the hardening is “soft” enough that the hardening approximates perfect plasticity, hardening plasticity can result in lack of conditioning. Usually, however, hardening is sufficiently stiff and therefore loss of ellipticity is not a concern.

⁷De Borst, R., Crisfield, M. A., Remmers, J. J., & Verhoosel, C. V. (2012). Nonlinear finite element analysis of solids and structures. John Wiley & Sons, at page 50–51.

(slip). The simplest evolution for α can be written as

$$\dot{\alpha} = |\dot{\varepsilon}^P|. \quad (2.84)$$

The **plastic flow rule**,

$$\dot{\varepsilon}^P = \dot{\gamma} \text{sign}(\sigma), \quad (2.85)$$

remains unchanged. The **Kuhn-Tucker conditions** are rewritten to include dependence on α ,

$$\dot{\gamma} \geq 0 \quad \Phi(\sigma, \alpha) \leq 0 \quad \dot{\gamma} \Phi(\sigma, \alpha) = 0, \quad (2.86)$$

and the **persistence condition** is written as

$$\dot{\gamma} \dot{\Phi}(\sigma, \alpha) = 0. \quad (2.87)$$

Determination of Plastic Multiplier

Like perfect plasticity, strain-hardening plasticity uses the time derivative of the yield function to generate an expression for the plastic multiplier $\dot{\gamma}$, which ultimately leads to a relationship between stress rates and strain rates, and the elasto-plastic tangent.

With the yield function in (2.83),

$$\begin{aligned} \frac{\partial \Phi}{\partial t} &= \frac{\partial \Phi}{\partial \sigma} \frac{\partial \sigma}{\partial t} + \frac{\partial \Phi}{\partial \alpha} \frac{\partial \alpha}{\partial t}, \\ &= \text{sign}(\sigma) \dot{\sigma} - H \dot{\alpha}, \\ &= \text{sign}(\sigma) E (\dot{\varepsilon} - \dot{\varepsilon}^P) - H |\dot{\varepsilon}^P|, \\ &= \text{sign}(\sigma) E \dot{\varepsilon} - E \dot{\gamma} - H \dot{\gamma} \stackrel{\text{set}}{=} 0, \end{aligned} \quad (2.88)$$

gives

$$\boxed{\dot{\gamma} = \dot{\varepsilon} \text{sign}(\sigma) \frac{E}{E + H}}. \quad (2.89)$$

The ratio $\frac{E}{E+H}$ in (2.89) may be conceived as the apportionment of total strain (rate) to plastic slip:

- If $H = 0$, then all strain is plastic slip, and the perfect plasticity results obtained in previous sections are obtained as a special case.
- If $H = E$, then half of all strain goes into plastic slip.
- If $H \gg E$, then very little $\dot{\varepsilon}$ goes into plastic slip.

Elasto-Plastic Tangent

The elasto-plastic tangent is found by observing $\dot{\sigma} = \frac{d\sigma}{d\varepsilon}\dot{\varepsilon}$. Starting with the elastic stress-strain relationship $\sigma = E(\varepsilon - \varepsilon^p)$, in the rate form and substituting for $\dot{\varepsilon}$ from (2.89) and for $\dot{\varepsilon}^p$ from the plastic flow rule (2.85) gives

$$\begin{aligned}\dot{\sigma} &= E(\dot{\varepsilon} - \dot{\varepsilon}^p), \\ &= \dot{\gamma} \operatorname{sign}(\sigma)(E + H) - \dot{\gamma} \operatorname{sign}(\sigma)E, \\ &= \dot{\gamma} \operatorname{sign}(\sigma)H.\end{aligned}\tag{2.90}$$

Substituting for $\dot{\gamma}$ from (2.89) gives

$$\dot{\sigma} = \frac{EH}{E + H}\dot{\varepsilon},\tag{2.91}$$

where E_{ep} , the elasto-plastic tangent, is giving by

$$\boxed{E_{ep} = \frac{EH}{E + H}}.\tag{2.92}$$

The concept of the elasto-plastic tangent is readily understood with the phenomenological model shown in Fig. 2.11.

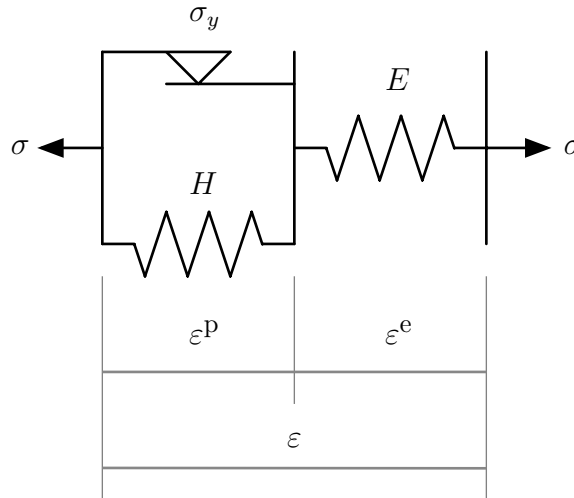


Figure 2.11: Phenomenological model of strain-hardening plasticity with applied stress σ , yield stress σ_y , hardening modulus H and elastic modulus E . The total strain, ε is the sum of plastic strain ε^p and elastic strain ε^e .

The stress σ is applied to the model. Initially, the applied stress σ creates only elastic strain ε^e since $\sigma \leq \sigma_y$. When $\sigma > \sigma_y$, strain composed of *both* elastic strain ε^e and plastic strain ε^p is created. The linear moduli E and H act as springs in series. The relationship for two springs with spring constants, k_1 and k_2 , in series immediately shows the relationship between the equivalent spring constant k_{eq} and the elasto-plastic tangent E_{ep}

$$\frac{1}{k_{eq}} = \frac{1}{k_1} + \frac{1}{k_2} \implies k_{eq} = \frac{k_1 k_2}{k_1 + k_2} \quad (2.93)$$

$$\begin{array}{ccc} \Downarrow & & \Downarrow \\ E_{ep} & = & \frac{EH}{E + H} \end{array} \quad (2.94)$$

The hardening modulus H is shown in Fig. 2.12(b) such that, during plastic flow, the incremental stress $\partial\sigma$ relative to the incremental plastic strain $\partial\varepsilon^p$ is given by

$$\frac{\partial\sigma}{\partial\varepsilon^p} = H. \quad (2.95)$$

2.9.3 Elasto-Plastic 2D Truss Element

Consider the two-dimensional truss member shown in Fig. 2.13. The coordinates in the reference configuration are noted with $\mathbf{X}^1$ and $\mathbf{X}^2$ for nodes $\dot{1}$ and $\dot{2}$, respectively. These nodes are moved into the current configuration at $\mathbf{x}^1$ and $\mathbf{x}^2$ through nodal displacements $\mathbf{u}^1$ and $\mathbf{u}^2$.

The strain energy of the truss is the strain energy density integrated over the truss's volume,

$$\mathcal{W} = \int_{\Omega} \boldsymbol{\sigma} : \boldsymbol{\varepsilon} \, d\Omega. \quad (2.96)$$

The virtual work done by the truss is

$$\delta\mathcal{W} = \int_{\Omega} \delta\boldsymbol{\varepsilon} : \boldsymbol{\sigma} \, d\Omega. \quad (2.97)$$

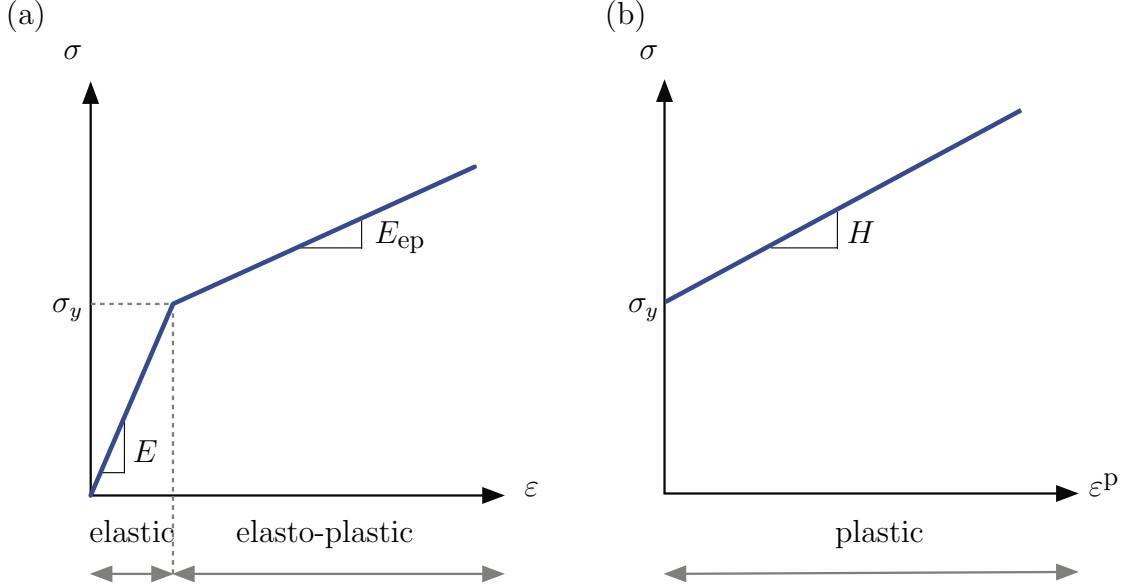


Figure 2.12: (a) Strain-hardening stress-strain relationship with elastic and elasto-plastic strain regions delineated, elastic modulus E , elasto-plastic modulus E_{ep} , and yield stress σ_y , and (b) stress-(plastic) strain relationship with hardening modulus H .

The virtual work computed in the reference configuration is then

$$\delta \mathcal{W} = \int_L \delta \boldsymbol{\epsilon} : \boldsymbol{\sigma} A \, dL, \quad (2.98)$$

$$\stackrel{(1D)}{=} \delta \left(\frac{\ell - L}{L} \right) \sigma A L = \frac{\delta \ell}{L} \sigma A L = \delta \ell \sigma A, \quad (2.99)$$

$$\stackrel{(2.70)}{=} (-\delta \mathbf{u}^1 + \delta \mathbf{u}^2) \cdot \hat{\mathbf{t}} \sigma A, \quad (2.100)$$

$$\stackrel{(\text{lin})}{=} (-\delta \mathbf{u}^1 + \delta \mathbf{u}^2) \cdot \hat{\mathbf{t}} E \left(\frac{\ell - L}{L} \right) A, \quad (2.101)$$

$$= (-\delta \mathbf{u}^1 + \delta \mathbf{u}^2) \cdot \hat{\mathbf{t}} k(\ell - L), \quad (2.102)$$

where the result from Hovey⁸ page 45, Eq. (2.70), $\delta \ell = (-\delta \mathbf{u}^1 + \delta \mathbf{u}^2) \cdot \hat{\mathbf{t}}$, has been used, and agrees with Eq. (2.76) of the same manuscript. The virtual work expressed

⁸Hovey CB (2000). Stress and Motion in Diarthrodial Joints Using Coupled Nonlinear Elastodynamics, Rigid Body Dynamics, and Large-Slip Contact-Impact, Doctoral Thesis, Stanford University, Stanford, CA.

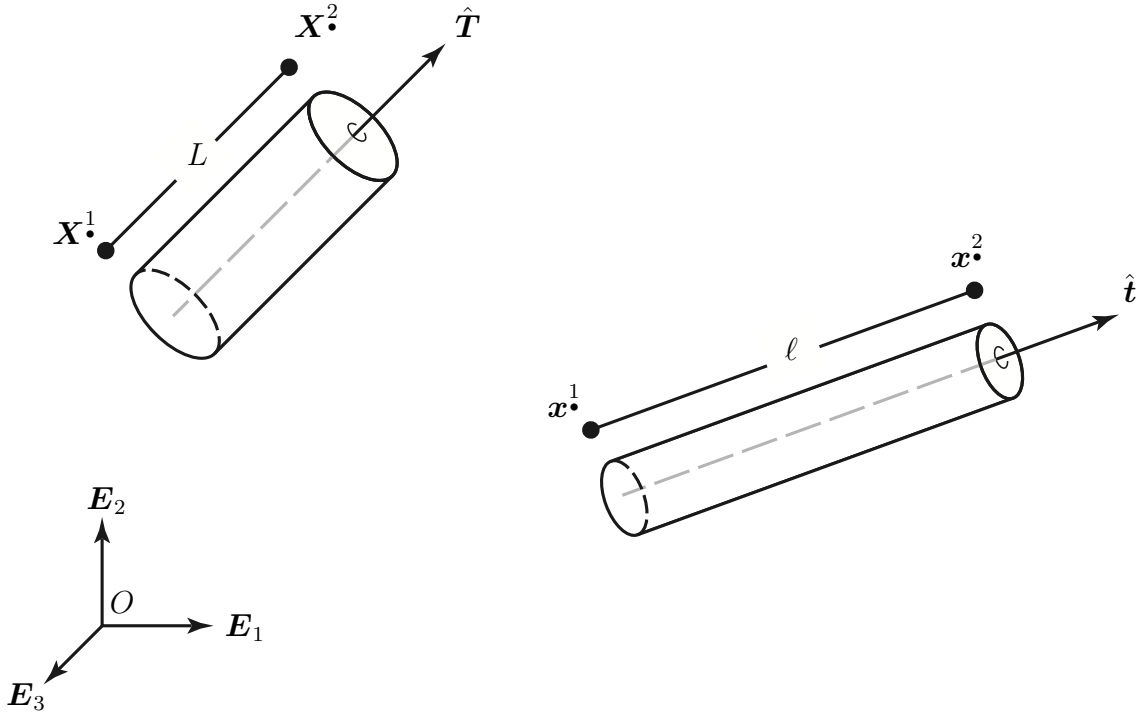


Figure 2.13: Nonlinear truss in reference and current configurations.

in terms of nodal displacements and the *nonlinear* internal force vector in (2.100) is then

$$\delta \mathcal{W} = \boldsymbol{\delta} \cdot \mathbf{f}^{\text{int}} = \left\{ \begin{array}{c} \delta \mathbf{u}^1 \\ \delta \mathbf{u}^2 \end{array} \right\}^T \sigma A \left\{ \begin{array}{c} -\hat{\mathbf{t}} \\ \hat{\mathbf{t}} \end{array} \right\}. \quad (2.103)$$

With $\delta \ell$ at hand, and noting that

$$\delta \hat{\mathbf{t}} = \delta \left(\frac{\mathbf{x}_2 - \mathbf{x}_1}{l} \right) = \delta \left((\mathbf{x}_2 - \mathbf{x}_1) l^{-1} \right), \quad (2.104)$$

$$= \frac{1}{l} [\mathbf{1} - \hat{\mathbf{t}} \otimes \hat{\mathbf{t}}] (-\delta \mathbf{u}_1 + \delta \mathbf{u}_2), \quad (2.105)$$

the second variation of the nonlinear form in (2.100), or equivalently in (2.103), is found through

$$\Delta(\delta \mathcal{W}) = \Delta \left[(-\delta \mathbf{u}^1 + \delta \mathbf{u}^2) \cdot \sigma A \hat{\mathbf{t}} \right], \quad (2.106)$$

$$\begin{aligned} &= (-\delta \mathbf{u}^1 + \delta \mathbf{u}^2) \cdot \Delta \sigma A \hat{\mathbf{t}} \\ &\quad + (-\delta \mathbf{u}^1 + \delta \mathbf{u}^2) \cdot \sigma A \Delta \hat{\mathbf{t}}. \end{aligned} \quad (2.107)$$

Now, adopting a compact notation where $\delta \mathbf{u} = (-\delta \mathbf{u}^1 + \delta \mathbf{u}^2)$ and $\Delta \mathbf{u} = (-\Delta \mathbf{u}^1 + \Delta \mathbf{u}^2)$,

$$\begin{aligned} \Delta(\delta \mathcal{W}) &= \delta \mathbf{u} \cdot \left\{ \frac{\partial \sigma}{\partial \varepsilon} \Delta \varepsilon A \hat{\mathbf{t}} + \sigma A \frac{1}{\ell} [\mathbf{1} - \hat{\mathbf{t}} \otimes \hat{\mathbf{t}}] \Delta \mathbf{u} \right\}, \end{aligned} \quad (2.108)$$

$$= \delta \mathbf{u} \cdot \left\{ \frac{\partial \sigma}{\partial \varepsilon} \left(\frac{\Delta \ell}{L} \right) A \hat{\mathbf{t}} + \frac{\sigma A}{\ell} [\mathbf{1} - \hat{\mathbf{t}} \otimes \hat{\mathbf{t}}] \Delta \mathbf{u} \right\}. \quad (2.109)$$

Expanding $\delta \mathbf{u}$ and $\Delta \mathbf{u}$ gives

$$\begin{aligned} \Delta(\delta \mathcal{W}) &= (-\delta \mathbf{u}^1 + \delta \mathbf{u}^2) \\ &\quad \cdot \left[\frac{\partial \sigma}{\partial \varepsilon} \frac{A}{L} \hat{\mathbf{t}} \otimes \hat{\mathbf{t}} + \frac{\sigma A}{\ell} [\mathbf{1} - \hat{\mathbf{t}} \otimes \hat{\mathbf{t}}] \right] \\ &\quad \left\{ (-\Delta \mathbf{u}^1 + \Delta \mathbf{u}^2) \right\}. \end{aligned} \quad (2.110)$$

This leads to the element stiffness equations

$$\Delta \delta \mathcal{W} = \boxed{\delta \cdot \mathbf{K} \Delta = \begin{Bmatrix} \delta \mathbf{u}^1 \\ \delta \mathbf{u}^2 \end{Bmatrix}^T \begin{bmatrix} \boldsymbol{\kappa} & -\boldsymbol{\kappa} \\ -\boldsymbol{\kappa} & \boldsymbol{\kappa} \end{bmatrix} \begin{Bmatrix} \Delta \mathbf{u}^1 \\ \Delta \mathbf{u}^2 \end{Bmatrix}}, \quad (2.111)$$

where

$$\boldsymbol{\kappa} = \left[\frac{\partial \sigma}{\partial \varepsilon} \frac{A}{L} \hat{\mathbf{t}} \otimes \hat{\mathbf{t}} + \frac{\sigma A}{\ell} [\mathbf{1} - \hat{\mathbf{t}} \otimes \hat{\mathbf{t}}] \right]. \quad (2.112)$$

Note that the explicit form of $\partial \sigma / \partial \varepsilon$ will depend on the plastic multiplier $\dot{\gamma}$. If $\dot{\gamma} = 0$, then $\partial \sigma / \partial \varepsilon = E$, as shown in the elastic region of Fig. 2.12 (a). If $\dot{\gamma} > 0$, then $\partial \sigma / \partial \varepsilon = E_{\text{ep}}$, as shown in the elasto-plastic region of Fig. 2.12 (a).

2.9.4 From 1D to 3D

In one-dimensional plasticity, we spoke of the Cauchy stress scalar $\sigma \in \mathbb{R}$ with positive values describing tension (resulting in elongations) and compression (resulting in contractions). Now, we generalize this concept to speak of the Cauchy stress tensor $\boldsymbol{\sigma}$ as a collection of all the normal and shear Cauchy stress components present in three dimensions.

We decompose $\boldsymbol{\sigma}$ into volumetric and deviatoric components, denoted

$$\boldsymbol{\sigma} = \boldsymbol{\sigma}_{\text{VOL}} + \boldsymbol{\sigma}_{\text{DEV}}. \quad (2.113)$$

Next, we *define* each of these two components. The **volumetric** Cauchy stress tensor is defined as

$$\boldsymbol{\sigma}_{\text{VOL}} \triangleq \frac{1}{3} \text{tr}(\boldsymbol{\sigma}) \mathbf{1} = \frac{1}{3} \sigma_{kk} \delta_{ij}. \quad (2.114)$$

It is often convenient to define **pressure** $P \in \mathbb{R}$ as

$$P \triangleq -\frac{1}{3} \text{tr}(\boldsymbol{\sigma}) \quad \begin{cases} P > 0 \text{ compression,} \\ P < 0 \text{ tension.} \end{cases} \quad (2.115)$$

Note the negative sign in the definition. The negative sign allows us to harmonize our concepts of (compressive, e.g., underwater) pressure as a positive, scalar with tension/compression (positive/negative) convention of the Cauchy stress tensor.

Furthermore, the deviatoric Cauchy stress tensor is often denoted simply as $\mathbf{s}$. Using this notation, the **deviatoric** Cauchy stress tensor is defined as

$$\mathbf{s} \triangleq \boldsymbol{\sigma} - \boldsymbol{\sigma}_{\text{VOL}} = \boldsymbol{\sigma} + P \mathbf{1}. \quad (2.116)$$

Uniaxial Tension Test

With these definitions at hand, we are now poised to demonstrate how the results of a uniaxial tension test at the plastic limit are used to generalize a theory of small-strain plasticity to three dimensions.

A yielding, the Cauchy stress tensor $\boldsymbol{\sigma}$ has the simple form

$$\boldsymbol{\sigma}|_y = \begin{bmatrix} \sigma_y & 0 & 0 \\ & 0 & 0 \\ & & 0 \end{bmatrix}. \quad (2.117)$$

Remark 2.9.1. Note that for symmetric matrices, we will avoid filling in the lower part of the matrix both for brevity, and to reinforce the idea of symmetry.

The volumetric part of the Cauchy stress tensor at yield is then

$$\boldsymbol{\sigma}_{\text{VOL}}|_y = \frac{\sigma_y}{3} \begin{bmatrix} 1 & 0 & 0 \\ & 1 & 0 \\ & & 1 \end{bmatrix}. \quad (2.118)$$

The deviatoric part of the Cauchy stress tensor at yield is then

$$\mathbf{s}|_y = \frac{3\sigma_y}{3} \begin{bmatrix} 1 & 0 & 0 \\ & 1 & 0 \\ & & 1 \end{bmatrix} - \frac{\sigma_y}{3} \begin{bmatrix} 1 & 0 & 0 \\ & 1 & 0 \\ & & 1 \end{bmatrix} = \frac{\sigma_y}{3} \begin{bmatrix} 2 & 0 & 0 \\ & -1 & 0 \\ & & -1 \end{bmatrix}. \quad (2.119)$$

So, we now have the form of the deviatoric Cauchy stress tensor $\mathbf{s}$ at yield. How can this result for a uniaxial tension test be generalized for three dimensions? The thought experiment is to envision $\mathbf{s}$ for *any and all* matrix forms, not just the form obtained in (2.119). Any and all matrix form for $\mathbf{s}$ will have one commonality: their matrix invariants.

Recall that the three invariants, $I_1(\mathbf{A})$, $I_2(\mathbf{A})$, and $I_3(\mathbf{A})$ of matrix $\mathbf{A}$ satisfy the equation

$$\det(\mathbf{A} - \lambda \mathbf{1}) \stackrel{\text{set}}{=} 0 \quad (2.120)$$

for any λ , and produce the characteristic equation

$$0 = \lambda^3 (-1) \quad (2.121)$$

$$+ \lambda^2 \underbrace{(a_{11} + a_{22} + a_{33})}_{\text{tr}(\mathbf{A})} \quad (2.122)$$

$$+ \lambda^1 \underbrace{(-a_{11}a_{22} - a_{22}a_{33} - a_{33}a_{11} + a_{12}a_{21} + a_{23}a_{32} + a_{31}a_{13})}_{\frac{1}{2} [\text{tr}(\mathbf{A} \cdot \mathbf{A}) - \text{tr}(\mathbf{A})\text{tr}(\mathbf{A})]} \quad (2.123)$$

$$+ \lambda^0 \underbrace{(a_{11}a_{22}a_{33} + a_{12}a_{23}a_{31} + a_{13}a_{21}a_{32} - a_{11}a_{23}a_{32} - a_{12}a_{21}a_{33} - a_{13}a_{22}a_{31})}_{\det(\mathbf{A})}. \quad (2.124)$$

Then the first invariant of $\mathbf{s}$,

$$J_1(\mathbf{s}) \triangleq \text{tr}(\mathbf{s}) = s_{kk} = 0, \quad (2.125)$$

is uninteresting because it has zero value for all stress states of $\boldsymbol{\sigma}$. The second invariant of $\mathbf{s}$,

$$J_2(\mathbf{s}) \triangleq \frac{1}{2} (\text{tr}(\mathbf{s} \cdot \mathbf{s})) = \frac{1}{2} s_{ij} s_{ji} \quad (2.126)$$

is interesting because we arrive at a quadratic form of $\mathbf{s}$. Returning to the thought experiment, we can now envision of form of $\mathbf{s}$ by way of $J_2(\mathbf{s})$ as valid for all values

Cauchy stress tensors $\boldsymbol{\sigma}$ and with plasticity occurring when $J_2(\mathbf{s})$ reaches $J_2(\mathbf{s}|_y)$. Said equivalently, we can imagine a yield function, as defined for scalars in (2.71), recast now for the entire stress tensor $\boldsymbol{\sigma}$ as

$$\Phi(\boldsymbol{\sigma}) \triangleq J_2(\mathbf{s}(\boldsymbol{\sigma})) - J_2(\mathbf{s}|_y) \leq 0. \quad (2.127)$$

Now

$$J_2(\mathbf{s}|_y) = \frac{1}{2} (\sigma_y^2)/9 (4 + 1 + 1) = \sigma_y^2/3. \quad (2.128)$$

Thus

$$\Phi(\boldsymbol{\sigma}) \triangleq J_2(\mathbf{s}(\boldsymbol{\sigma})) - \frac{\sigma_y^2}{3} \leq 0. \quad (2.129)$$

Now, on the yield surface, $\Phi = 0$, which means

$$J_2(\mathbf{s}(\boldsymbol{\sigma}|_y)) = \frac{\sigma_y^2}{3} \implies \sqrt{3J_2(\mathbf{s}(\boldsymbol{\sigma}|_y))} = \sigma_y. \quad (2.130)$$

This gives rise to the form of the yield function of so-called J_2 **plasticity**,

$$\boxed{\Phi(\boldsymbol{\sigma}) \triangleq \sqrt{3J_2(\mathbf{s}(\boldsymbol{\sigma}))} - \sigma_y \leq 0.} \quad (2.131)$$

The quantity $\sqrt{3J_2}$ is often referred to as the **von Mises stress** or **equivalent tensile stress** and denoted as σ_{VM} . Thus, the yield function may be equivalently written as

$$\Phi(\boldsymbol{\sigma}) \triangleq \sigma_{\text{VM}}(\boldsymbol{\sigma}) - \sigma_y \leq 0. \quad (2.132)$$

2.10 Thermodynamics

2.10.1 Properties

Thermodynamics properties can be categorized as either **intensive** or **extensive**.

- Intensive properties are independent of, and therefore not proportional to, the size or sample of the material. Intensive properties are also referred to as **intrinsic** properties or, equivalently, as **bulk** properties.
- In contrast, extensive properties are dependent on, and therefore proportional to, the size or sample of the material. Extrinsic properties are also referred to as **extrinsic** properties.

Figure 2.14 demonstrates the independence of intrinsic properties and the dependence of extrinsic properties on the control volume, designated with the dashed lines. The original control volume, on the left side of the figure, is divided into two equal-sized control volumes, as shown on the right side of the figure. The intrinsic properties do not depend on the size of the control volume. The extrinsic properties, however, are now scaled by a factor of $\frac{1}{2}$.

Intrinsic (intensive, bulk) pressure, density, temperature	P, ρ, T	P, ρ, T	P, ρ, T
Extrinsic (extensive) mass, volume, energy	m, V, E	$\frac{m}{2}, \frac{V}{2}, \frac{E}{2}$	$\frac{m}{2}, \frac{V}{2}, \frac{E}{2}$
Specific (extrinsic/mass) specific volume, specific energy	ν, ε	ν, ε	ν, ε

Figure 2.14: Thermodynamic properties categorized as intrinsic, extrinsic, and specific.

It is often convenient to speak of extrinsic properties in a way that makes them *independent* of size or sample of the material. To achieve this, the extrinsic properties are divided by the mass of the sample. This ratio is referred to as the specific extrinsic property. Two common examples of specific properties are the **specific volume** and

specific energy, defined as

$$\mathcal{V} \triangleq \frac{V}{m} = \frac{1}{\rho}, \quad (2.133)$$

and

$$\mathcal{E} \triangleq \frac{E}{m}. \quad (2.134)$$

2.10.2 First Law of Thermodynamics

The first law for a closed system states the change in internal energy $\Delta\mathcal{I}$ is equal to the net energy Q added to the system as heat less the net work W done **by** the system.

$$\Delta\mathcal{I} = Q - W. \quad (2.135)$$

This was the sign convention used by Joule and Clausius. A different sign convention, used by Planck, stated that the change in internal energy $\Delta\mathcal{I}$ is equal to the net energy Q added to the system as heat plus the net work done **on** the system.

$$\Delta\mathcal{I} = Q + W. \quad (2.136)$$

Thus, all net transfers **to** the system are **positive**. In contrast, all net transfers **from** the system are **negative**.

Then, the conservation of energy principle for open systems can be written as

$$\Delta E = \Delta(\mathcal{K} + \mathcal{I} + \mathcal{P}) = Q + W, \quad (2.137)$$

where the total energy E is the sum of the kinetic energy $\mathcal{K}$, the internal energy $\mathcal{I}$, and the potential energy $\mathcal{P}$. The change in total energy is equal to the net energy Q added to the system as heat plus the net work W done **on** the system.

Next, we introduce

- $d(\cdot)$ as the **exact differential**, which does not depend on the path taken through space of the thermodynamic parameters and depends only on the initial states, and
- $\delta(\cdot)$ as the **inexact differential**, which depends on the path taken through space of thermodynamic parameters.

The balance law then states

$$dE = \delta Q + \delta W. \quad (2.138)$$

2.10.3 Second Law of Thermodynamics

The Second Law states that the infinitesimal increment of entropy dS of a system from an infinitesimal transfer of heat δQ , divided by the common temperature of the system and the surrounding that supplies the heat, satisfies

$$dS \geq \frac{\delta Q}{T}. \quad (2.139)$$

For a reversible process,

$$dS = \frac{\delta Q}{T}. \quad (2.140)$$

For an irreversible process,

$$dS > \frac{\delta Q}{T}. \quad (2.141)$$

2.10.4 First and Second Laws Combined

For a reversible process, the first and second laws can be combined as

$$\boxed{dE = TdS + \delta W, \quad (\text{1st and 2nd Law, Reversible Form}).} \quad (2.142)$$

2.10.5 Thermodynamic Potentials

Variables

P	pressure	(N/m ²)
V	volume	(m ³)
T	temperature	(K)
S	entropy	(J/K)

Potentials

E	(?internal only?) energy	(J)
$F \triangleq E - TS$	Helmholtz free energy	(J)
$G \triangleq H - TS$	Gibbs free energy	(J)
$H \triangleq E + PV$	enthalpy	(J)

The **Helmholtz** free energy F is the Legendre transform of the (?internal?) energy E , where the entropy S is replaced by the temperature T as the independent variable.

Differential Forms of Potentials

$$dE = T dS - P dV \quad \text{to come}$$

$$dF = -P dV - S dT \quad \text{xx}$$

$$dG = -S dT + V dP \quad \text{xx}$$

$$dH = V dP + T dS \quad \text{xx}$$

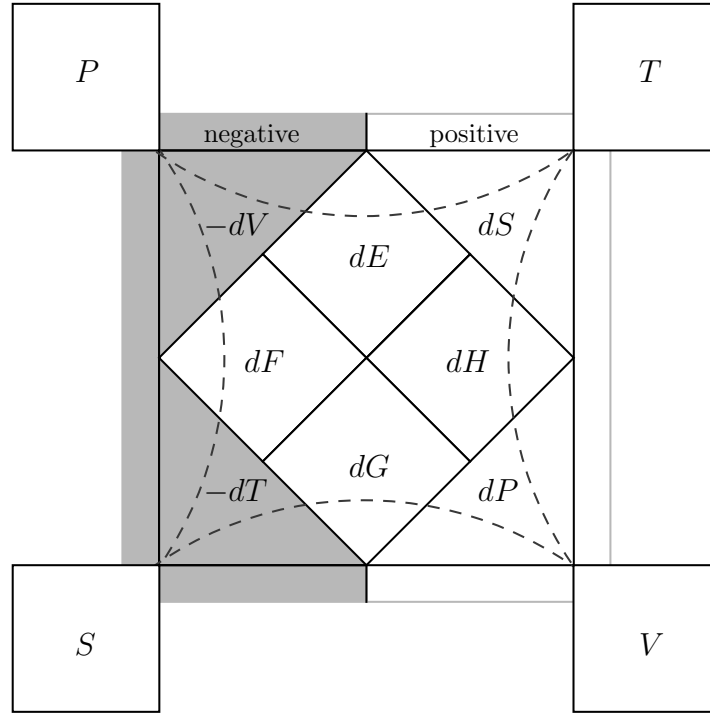


Figure 2.15: Mnemonic for the quick recall of the thermodynamic potentials in differential form.

2.10.6 Maxwell Relations

The Maxwell relations are a statement of equality between *first* derivative of a function because of a quality possessed by the *second* derivative of that function. Specifically, the Maxwell relations equate *first* derivatives of the thermodynamic potential functions E, F, G , and H , because the *second* derivatives of those same functions does not depend on the order of differentiation with respect to the thermodynamic variables P, V, T , and S .

The Maxwell relations make use of the **Schwarz theorem**, which states that if function $f : \mathbb{R}^2 \mapsto \mathbb{R}$ has continuous second partial derivatives at a given point $\mathbf{p} \in \mathbb{R}^2 = \{p_1, p_2\}^T$,⁹ then

$$\frac{\partial^2 f(\mathbf{p})}{\partial x_i \partial x_k} = \frac{\partial^2 f(\mathbf{p})}{\partial x_k \partial x_i} \quad (\text{Commutative}) \quad (2.143)$$

The Schwarz theorem establishes that the result of differentiation of a function of two variables is independent of the order of that differentiation.

With this context in mind, consider now the *first* derivatives of the thermodynamic potentials with respect to their natural variables. Specifically, we can write the four thermodynamic potentials as functions of their natural variables:

$$E = E(S, V), \quad (2.144)$$

$$F = F(V, T), \quad (2.145)$$

$$G = G(T, P), \quad (2.146)$$

$$H = H(P, S). \quad (2.147)$$

Next, we take *first* derivatives of these potentials:

$$\left(\frac{\partial E}{\partial S} \right)_V = T, \quad \left(\frac{\partial E}{\partial V} \right)_S = -P, \quad (2.148)$$

$$\left(\frac{\partial F}{\partial V} \right)_T = -P, \quad \left(\frac{\partial F}{\partial T} \right)_V = -S, \quad (2.149)$$

$$\left(\frac{\partial G}{\partial T} \right)_P = -S, \quad \left(\frac{\partial G}{\partial P} \right)_T = V, \quad (2.150)$$

$$\left(\frac{\partial H}{\partial P} \right)_S = V, \quad \left(\frac{\partial H}{\partial S} \right)_P = T. \quad (2.151)$$

Finally, we take *second* derivatives of the potential functions with respect to the natural variables, which allows for the *first* derivatives of the natural variables with respect to the natural variables themselves to be equated, as follows:

$$\frac{\partial^2 E}{\partial V \partial S} \implies \left(\frac{\partial T}{\partial V} \right)_S = \left(\frac{-\partial P}{\partial S} \right)_V \quad (2.152)$$

⁹This relationship can be extended to general dimensionality of $f : \mathbb{R}^n \mapsto \mathbb{R}$, but this higher dimensionality is not used in the development that follows.

$$\frac{\partial^2 F}{\partial T \partial V} \Rightarrow \left(\frac{-\partial P}{\partial T} \right)_V = \left(\frac{-\partial S}{\partial V} \right)_T \quad (2.153)$$

$$\frac{\partial^2 G}{\partial P \partial T} \Rightarrow \left(\frac{-\partial S}{\partial P} \right)_T = \left(\frac{\partial V}{\partial T} \right)_P \quad (2.154)$$

$$\frac{\partial^2 H}{\partial S \partial P} \Rightarrow \left(\frac{\partial V}{\partial S} \right)_P = \left(\frac{\partial T}{\partial P} \right)_S \quad (2.155)$$

The foregoing equations are useful for eliminating entropy S , which is difficult to measure in a laboratory setting, in favor of the other natural variables pressure P , volume V , and temperature T , all of which are relatively easier to measure.

2.11 Shock

2.11.1 Introduction

2.11.2 An Ideal Fluid

2.11.3 Definition of a Shock

2.11.4 Conservation Laws for a Single Shock

Laboratory Fixed Coordinate Frame

The derivation that follows assumes a **laboratory-fixed reference frame**,¹⁰ as shown in Fig. 2.16. Consider a rigid tube containing a rigid piston and an ideal fluid at ambient density ρ_0 and ambient pressure P_0 at time $t = t_0$.

Assume at time $t = (t_0 + \Delta t)$, the piston moves left-to-right along the x -axis with speed u_P , measured in the fixed laboratory reference frame. Assume the piston speed u_P causes a shock to propagate through the material with speed U_S . We require the following:

$$0 < u_P < +\infty \quad (2.156)$$

$$0 < U_S < +\infty \quad (2.157)$$

$$u_P < U_S \quad (2.158)$$

From these considerations, we conclude that the material to the left of the shock wave in Fig. 2.16 moves to the right at speed u_p . The material to the right of the shock

¹⁰A shock-fixed reference frame derivation is developed in a subsequent section.

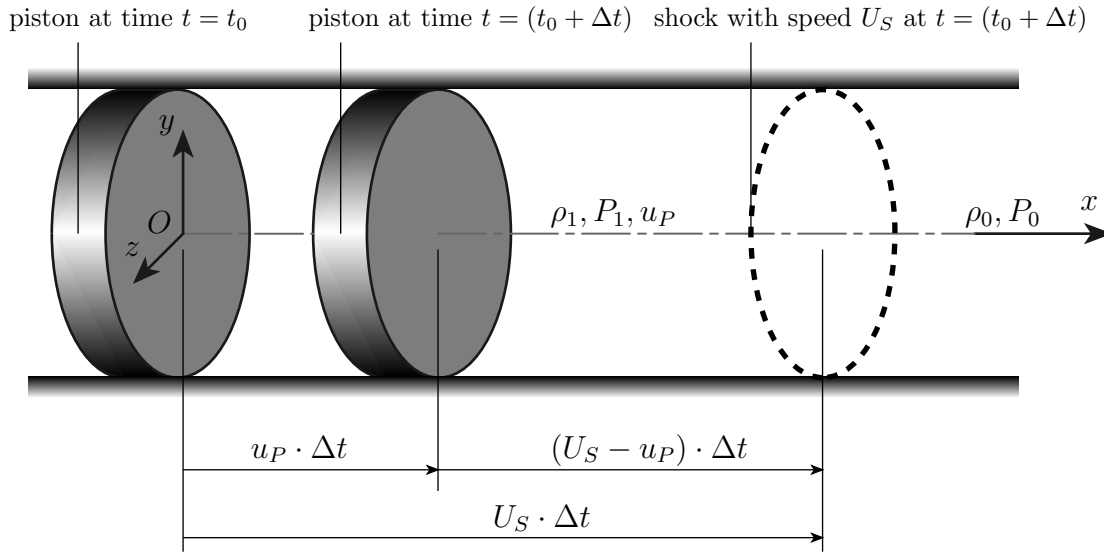


Figure 2.16: Piston moving left-to-right along the x -axis, with piston speed u_P , causing shock with shock speed U_S .

wave remains at rest.¹¹

Since the piston compresses the material in the tube, we know

- $\rho_1 > \rho_0$,

where ρ_1 is the density of the material to the left of the shock.

Allow some time interval Δt to elapse, as illustrated in Fig. 2.16. Consider the distances along the x -axis, written as a multiplication of speeds and elapsed time. In the time interval Δt , the piston moves along the x -axis a distance of $u_P \cdot \Delta t$. In the same time interval, the shock wave moves along the x -axis a distance of $U_S \cdot \Delta t$. Thus, the control volume at $t = (t_0 + \Delta t)$ has an axial length of $(U_S - u_P) \cdot \Delta t$.

From these axial lengths and piston cross-sectional area A , we can write the following relationships:

- volume

$$V_0 = V|_{t=t_0} = A \cdot U_S \cdot \Delta t,$$

¹¹Furthermore, we assume the material throughout the tube and for all time is inviscid, non-diffusive, and non-reacting. We assume an adiabatic boundary, thus no heat conduction occurs through the boundary of the tube walls or piston. We assume no heat transfer from radiation occurs across the shock. Finally, we assume body forces from gravity and electromagnetism are negligible.

$$V_1 = V|_{t=(t_0+\Delta t)} = A \cdot (U_S - u_P) \cdot \Delta t,$$

- density

$$\rho_0 = \rho|_{t=t_0},$$

$$\rho_1 = \rho|_{t=(t_0+\Delta t)},$$

- mass

$$\rho_0 V_0 = \rho_0 \cdot A \cdot U_S \cdot \Delta t,$$

$$\rho_1 V_1 = \rho_1 \cdot A \cdot (U_S - u_P) \cdot \Delta t.$$

Conservation of Mass

Conservation of mass within the control volume requires that mass at time $t = (t_0 + \Delta t)$ be unchanged from mass at time $t = t_0$. Thus

$$\rho_0 \cdot \mathcal{A} \cdot U_S \cdot \Delta t = \rho_1 \cdot \mathcal{A} \cdot (U_S - u_P) \cdot \Delta t, \quad (2.159)$$

or

$$\boxed{\rho_0 U_S = \rho_1 (U_S - u_P), \quad \text{(Conservation of Mass).}} \quad (2.160)$$

Using specific volume (2.133), conservation of mass be rewritten as

$$\mathcal{V}_1 = \mathcal{V}_0 \left(1 - \frac{u_P}{U_S} \right), \quad (2.161)$$

or

$$\boxed{\mathcal{V}_1 - \mathcal{V}_0 = \frac{-v_0 u_P}{U_S}, \quad \text{(Conservation of Mass, Volume Form).}} \quad (2.162)$$

From (2.156)–(2.158), we know $\mathcal{V}_1 < \mathcal{V}_0$, which makes sense since we are compressing the material and generally have the understanding that increased pressures causes decreased volumes.

Conservation of Momentum

The net driving force from the piston on the material is $(P_1 - P_0) \cdot A$. Conservation of momentum requires $F = d(ma)/dt$. Written over the time interval Δt and for constant mass, conservation of momentum can be written

$$F \cdot \Delta t = m \cdot \Delta U. \quad (2.163)$$

Rewriting this expression in terms of pressures and densities, we have

$$(P_1 - P_0) \cdot \mathcal{A} \cdot \Delta t = (\rho_0 \cdot \mathcal{A} \cdot U_S \cdot \Delta t) (u_P - 0) \quad (2.164)$$

which gives

$$\boxed{P_1 - P_0 = \rho_0 U_S u_P, \quad \text{(Conservation of Momentum).}} \quad (2.165)$$

Using (2.133), (2.165) can be rewritten as

$$\boxed{P_1 - P_0 = \frac{U_S u_P}{V_0}, \quad \text{(Conservation of Momentum, Volume Form).}} \quad (2.166)$$

Conservation of Energy

The work done on the control volume by the piston moving left-to-right is given by

$$W_P = \underbrace{P_1 \cdot A}_{\text{force}} \cdot \underbrace{u_P \cdot \Delta t}_{\text{disp.}} \quad (2.167)$$

Note that the right boundary of the control volume, located at the shock location, does not move, so the work across this boundary is identically zero.

The kinetic energy increase of the control volume is simply the one-half times the mass times the square of the change in velocity, thus

$$KE = \frac{1}{2} \underbrace{(\rho_0 \cdot A \cdot U_S \cdot \Delta t)}_{\text{mass}} \underbrace{(u_P - 0)^2}_{\text{velocity}} \quad (2.168)$$

Finally, the increase in internal energy is just the internal energy **per unit mass**, $\mathcal{E}$, multiplied by the mass,

$$IE = (\mathcal{E}_1 - \mathcal{E}_0) \underbrace{(\rho_0 \cdot A \cdot U_S \cdot \Delta t)}_{\text{mass}} \quad (2.169)$$

Equating these quantities gives

$$P_1 \cdot \mathcal{A} \cdot u_P \cdot \Delta t = \frac{1}{2} \rho_0 \cdot \mathcal{A} \cdot U_S \cdot \Delta t \cdot u_P^2 + (\mathcal{E}_1 - \mathcal{E}_0) \rho_0 \cdot \mathcal{A} \cdot U_S \cdot \Delta t \quad (2.170)$$

or simply

$$\boxed{P_1 u_P = \rho_0 U_S \left(\frac{u_P^2}{2} + \mathcal{E}_1 - \mathcal{E}_0 \right), \quad \text{(Conservation of Energy).}} \quad (2.171)$$

The foregoing equation can be written without kinematic variables U_S and u_P . Next, note that conservation of mass can be rewritten as

$$\frac{u_P}{U_S} = 1 - \frac{\rho_0}{\rho_1}. \quad (2.172)$$

Then, with (2.171),

$$P_1 u_P \stackrel{(2.165)}{=} (P_1 - P_0) \frac{u_P}{2} + \rho_0 U_S (\mathcal{E}_1 - \mathcal{E}_0), \quad (2.173)$$

$$= -P_0 u_P + 2 \rho_0 U_S (\mathcal{E}_1 - \mathcal{E}_0), \quad (2.174)$$

$$(\mathcal{E}_1 - \mathcal{E}_0) = \frac{u_P (P_1 + P_0)}{2 \rho_0 U_S}, \quad (2.175)$$

$$\stackrel{(2.172)}{=} \left(\frac{P_1 + P_0}{2 \rho_0} \right) \left(1 - \frac{\rho_0}{\rho_1} \right), \quad (2.176)$$

$$\stackrel{(2.133)}{=} \left(\frac{P_1 + P_0}{2 \mathcal{V}_0} \right) \left(1 - \frac{\mathcal{V}_1}{\mathcal{V}_0} \right), \quad (2.177)$$

which simplifies to

$$\boxed{\mathcal{E}_1 - \mathcal{E}_0 = \frac{-1}{2} (P_1 + P_0) (\mathcal{V}_1 - \mathcal{V}_0), \quad \textbf{(Rankine-Hugoniot curve).}} \quad (2.178)$$

The Rankine-Hugoniot equation is the Conservation of Energy jump condition with state variables alone; there are no kinematic variables in the equation. To plot the Rankine-Hugoniot relationship in the $P\mathcal{V}$ plane with $\mathcal{E}_1 > \mathcal{E}_0$, the initial state $(\mathcal{V}_0, P_0)$ must be known.

Speed-Volume-Pressure Relationships

The alternative forms for mass conservation (2.162) and momentum conservation (2.166), when multiplied together or divided by one another, readily allow for the shock speed U_S and particle speed u_P to be written in terms of volume and pressure. These relationship are found to be

$$U_S = \mathcal{V}_0 \sqrt{\frac{P_1 - P_0}{-(\mathcal{V}_1 - \mathcal{V}_0)}} = \mathcal{V}_0 \sqrt{\frac{\Delta P}{-\Delta \mathcal{V}}}, \quad (2.179)$$

and

$$u_P = \sqrt{-(P_1 - P_0) (\mathcal{V}_1 - \mathcal{V}_0)} = \sqrt{-\Delta P \Delta \mathcal{V}}. \quad (2.180)$$

Alternative Formulations

Now we develop the shock-fixed reference frame equations for mass, momentum, and energy conservation. (To come.)

2.11.5 $P\mathcal{V}$ Diagram

The $P\mathcal{V}$ diagram shows pressure P as a function of specific volume $\mathcal{V}$. Thus, the state variables P and $\mathcal{V}$ are considered in isolation from the other state and kinematic variables. The initial laboratory state is denoted $(\mathcal{V}_0, P_0)$. A large compressive pressure P_1 is applied to the material, which causes the volume to shrink to $\mathcal{V}_1$. If this applied pressure P_1 causes a shock, we record the pressure and volume state, and then repeat the experiment at a different pressure, e.g., P_2 . Several compression experiments are run, and the $P\mathcal{V}$ states that created a shock in the material are recorded in the $P\mathcal{V}$ diagram.

For concreteness, we show four experiments with applied pressures $P_1 \dots P_4$ and resultant volumes $\mathcal{V}_1 \dots \mathcal{V}_4$, shown in Fig. 2.17. Note that all experiments start from the same initial pressure P_0 and initial specific volume $\mathcal{V}_0$.

These four thermodynamic states lie on a locus of all possible thermodynamic states that can exist in the material behind a shock. This locus is referred to as the **shock Hugoniot**. The curve connecting the locus is the **Rankine-Hugoniot curve**. The line connecting two end states is the **Rayleigh line**.

It is important to emphasize that the Rankine-Hugoniot curve **is not** the path taken by the material between two thermodynamic states.

Note that (2.179) describes the mathematical increase in shock speed with increasing slope of the Hugoniot, indicated in Fig. 2.17. Specifically, the slope of the Hugoniot goes as the negative squared value of the shock speed U_S per initial specific volume $\mathcal{V}_0$, or

$$\frac{\Delta P}{\Delta \mathcal{V}} = - \left(\frac{U_S}{\mathcal{V}_0} \right)^2. \quad (2.181)$$

2.11.6 Pu_P Diagram

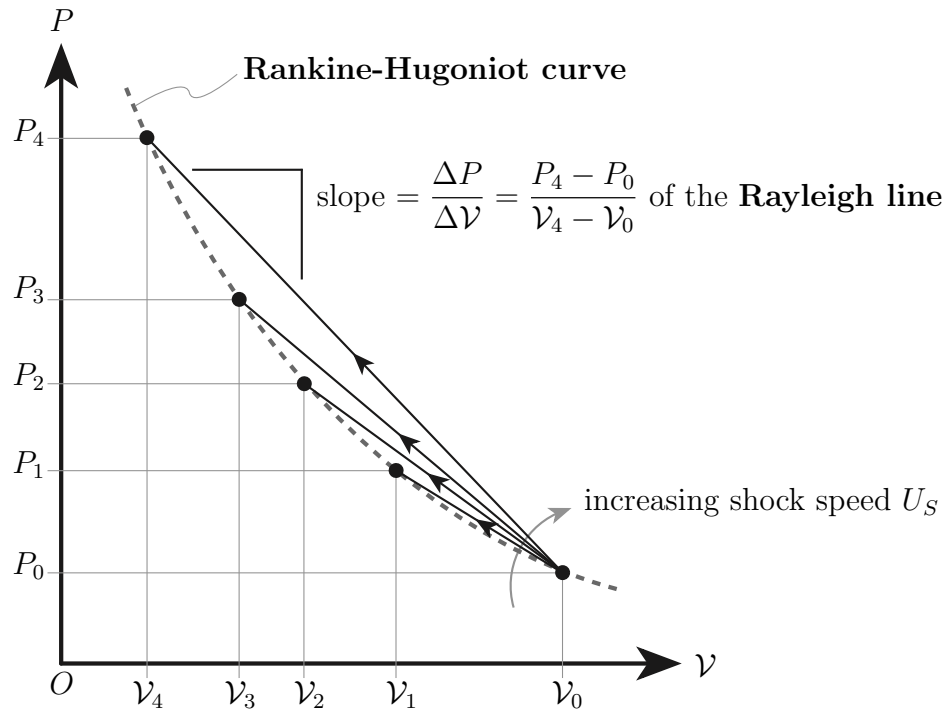


Figure 2.17: Pressure as a function of volume for the shock Hugoniot locus of thermodynamic equilibrium states.

Chapter 3

Implementation

3.1 Finite Elements

3.1.1 Lines

LL2

The linear two-noded line finite element has shape functions with the general form $\phi(\xi) = A + B\xi$.

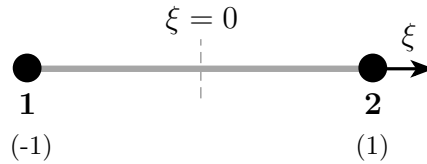


Figure 3.1: Linear two-noded line finite element.

$$\phi_1(\xi) = -\frac{1}{2} \cdot (\xi - 1) \quad (3.1)$$

$$\phi_2(\xi) = \frac{1}{2} (\xi + 1) \cdot \quad (3.2)$$

Remark 3.1.2. Note that “ $\cdot$ ” as used here has a dual purpose. First, it denotes multiplication, and by this measure alone, is unnecessary to include. However, the second purpose is a connotation of location, from left-to-right on the ξ -axis, where the shape function must be unity. Moreover, the quantities in parentheses () connote where the shape function must go to zero.

LL3

The linear three-noded line finite element has shape functions with the general form $\phi(\xi) = A + B\xi + C\xi^2$.

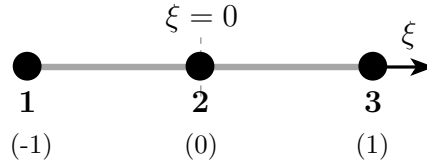


Figure 3.2: Linear three-noded line finite element.

$$\phi_1(\xi) = \frac{1}{2} \cdot (\xi) \cdot (\xi - 1) \quad (3.3)$$

$$\phi_2(\xi) = -1 \cdot (\xi + 1) \cdot (\xi - 1) \quad (3.4)$$

$$\phi_3(\xi) = \frac{1}{2} \cdot (\xi + 1) \cdot (\xi) \cdot \quad (3.5)$$

See Remark 3.1.1 for an explanation of the multiplication “ $\cdot$ ” and parentheses “ $(\quad)$ ” notation.

LL4

The linear four-noded line finite element has shape functions with the general form $\phi(\xi) = A + B\xi + C\xi^2 + D\xi^3$.

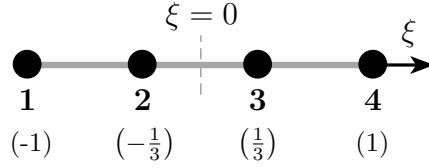


Figure 3.3: Linear four-noded line finite element.

$$\phi_1(\xi) = \frac{-9}{16} \cdot \left(\xi + \frac{1}{3} \right) \left(\xi - \frac{1}{3} \right) (\xi - 1) \quad (3.6)$$

$$\phi_2(\xi) = \frac{9}{16} (\xi + 1) \cdot \left(\xi - \frac{1}{3} \right) (\xi - 1) \quad (3.7)$$

$$\phi_3(\xi) = \frac{-9}{16} (\xi + 1) \left(\xi + \frac{1}{3} \right) \cdot (\xi - 1) \quad (3.8)$$

$$\phi_4(\xi) = \frac{9}{16} (\xi + 1) \left(\xi + \frac{1}{3} \right) \left(\xi - \frac{1}{3} \right) \cdot \quad (3.9)$$

See Remark 3.1.1 for an explanation of the multiplication “ $\cdot$ ” and parentheses () notation.

3.1.2 Triangles

LT3

The linear three-noded triangular finite element has shape functions with the general form $\phi(\boldsymbol{\xi}) = A + B\xi_1 + C\xi_2$.

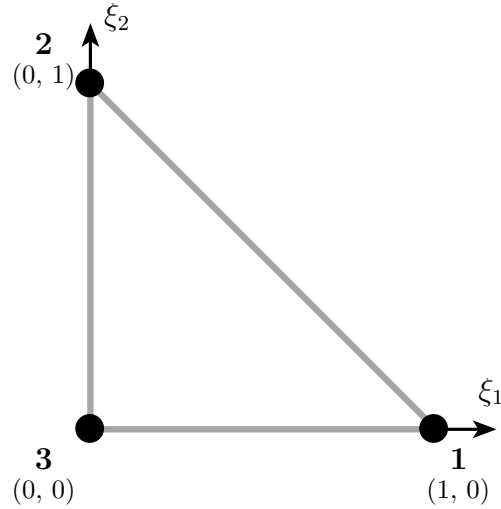


Figure 3.4: Linear three-noded triangular finite element.

$$\phi_1(\boldsymbol{\xi}) = \xi_1 \tag{3.10}$$

$$\phi_2(\boldsymbol{\xi}) = \xi_2 \tag{3.11}$$

$$\phi_3(\boldsymbol{\xi}) = 1 - \xi_1 - \xi_2 \tag{3.12}$$

LT6

The linear six-noded triangular finite element has shape functions with the general form $\phi(\boldsymbol{\xi}) = A + B\xi_1 + C\xi_2 + D\xi_1\xi_2 + E\xi_1^2\xi_2 + F\xi_1\xi_2^2$.

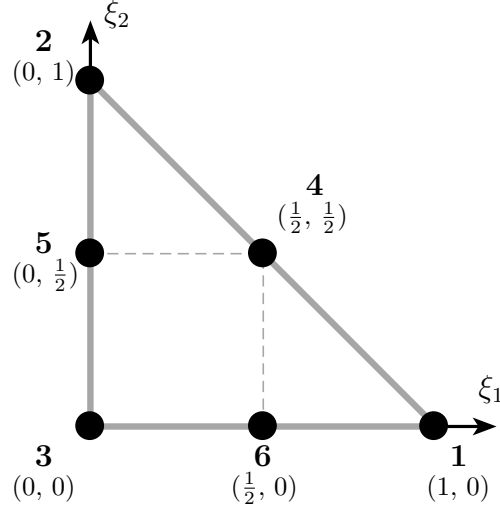


Figure 3.5: Linear six-noded triangular finite element.

$$\phi_1(\boldsymbol{\xi}) = 2 \xi_1 \left(\xi_1 - \frac{1}{2} \right) \quad (3.13)$$

$$\phi_2(\boldsymbol{\xi}) = 2 \xi_2 \left(\xi_2 - \frac{1}{2} \right) \quad (3.14)$$

$$\phi_3(\boldsymbol{\xi}) = 2 (1 - \xi_1 - \xi_2) \left(\frac{1}{2} - \xi_1 - \xi_2 \right) \quad (3.15)$$

$$\phi_4(\boldsymbol{\xi}) = 4 \xi_1 \xi_2 \quad (3.16)$$

$$\phi_5(\boldsymbol{\xi}) = 4 (1 - \xi_1 - \xi_2) \xi_2 \quad (3.17)$$

$$\phi_6(\boldsymbol{\xi}) = 4 (1 - \xi_1 - \xi_2) \xi_1 \quad (3.18)$$

LT6 Detailed Derivation

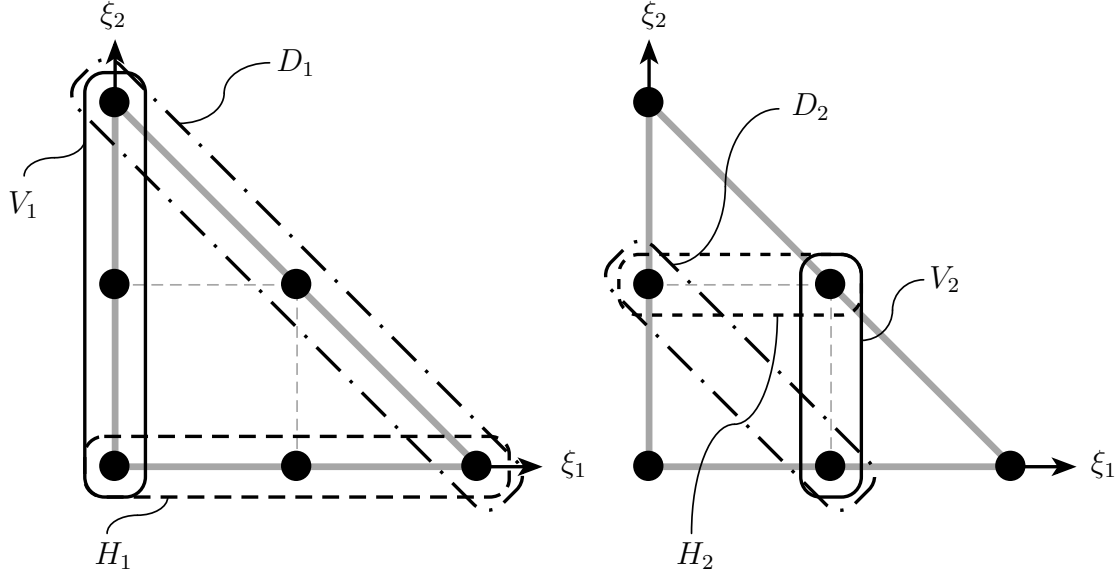


Figure 3.6: Detailed derivation of the linear six-noded triangular finite element, with horizontal lines H_1 and H_2 , vertical lines V_1 and V_2 , and diagonal lines D_1 and D_2 .

We begin by writing equations of two horizontal, two vertical, and two diagonal lines, as outlined in Fig. (3.6).

$$H_1 : \xi_2 = 0 \quad \forall \quad \xi_1 \tag{3.19}$$

$$H_2 : \xi_2 = \frac{1}{2} \quad \forall \quad \xi_1 \implies 0 = \left(\xi_2 - \frac{1}{2} \right) \tag{3.20}$$

$$V_1 : \xi_1 = 0 \quad \forall \quad \xi_2 \tag{3.21}$$

$$V_2 : \xi_1 = \frac{1}{2} \quad \forall \quad \xi_2 \implies 0 = \left(\xi_1 - \frac{1}{2} \right) \tag{3.22}$$

$$D_1 : \xi_2 = 1 - \xi_1 \implies 0 = (1 - \xi_1 - \xi_2) \tag{3.23}$$

$$D_2 : \xi_2 = \frac{1}{2} - \xi_1 \implies 0 = \left(\frac{1}{2} - \xi_1 - \xi_2 \right) \tag{3.24}$$

These six equations represent lines where the shape function has zero value for all combinations of (ξ_1, ξ_2) . Using these lines, we can construct shape functions as a functional products of (ξ_1, ξ_2) and locations (e.g., lines that intersect nodes) where the nodal values of the shape function must be identically zero.

Let shape function ϕ_1 have coefficient A_1 . Then ϕ_1 can be written as

$$\phi_1(\boldsymbol{\xi}) = A_1 \cdot V_1 \cdot V_2 = A_1 \cdot \xi_1 \cdot \left(\xi_1 - \frac{1}{2} \right). \quad (3.25)$$

Then, evaluating ϕ_1 at the node **1** location gives

$$\phi_1(1, 0) = A_1(1) \left(\frac{1}{2} \right) \stackrel{\text{set}}{=} 1 \implies A_1 = 2. \quad (3.26)$$

The form of ϕ_2 follows similarly to ϕ_1 . Next, consider ϕ_3 .

$$\phi_3(\boldsymbol{\xi}) = A_3 \cdot D_1 \cdot D_2 = A_3 \cdot \left(\frac{1}{2} - \xi_1 - \xi_2 \right) \cdot (1 - \xi_1 - \xi_2). \quad (3.27)$$

Then, evaluating ϕ_3 at the node **3** location gives

$$\phi_3(0, 0) = A_3 \left(\frac{1}{2} \right) (1) \stackrel{\text{set}}{=} 1 \implies A_3 = 2. \quad (3.28)$$

Finally, ϕ_4 , ϕ_5 and ϕ_6 can be found readily (we omit some details now that the pattern has been shown above) as

$$\phi_4 = A_4 \cdot H_1 \cdot V_1 \quad \text{and} \quad A_4 = 4, \quad (3.29)$$

$$\phi_5 = A_5 \cdot H_1 \cdot D_1 \quad \text{and} \quad A_5 = 4, \quad (3.30)$$

$$\phi_6 = A_6 \cdot V_1 \cdot D_1 \quad \text{and} \quad A_6 = 4. \quad (3.31)$$

LT10

The linear ten-noded triangular finite element has shape functions with the general form $\phi(\boldsymbol{\xi}) = A + B\xi_1 + C\xi_2 + D\xi_1\xi_2 + E\xi_1^2 + F\xi_2^2 + G\xi_1^2\xi_2 + H\xi_1\xi_2^2 + I\xi_1^3 + J\xi_2^3$.

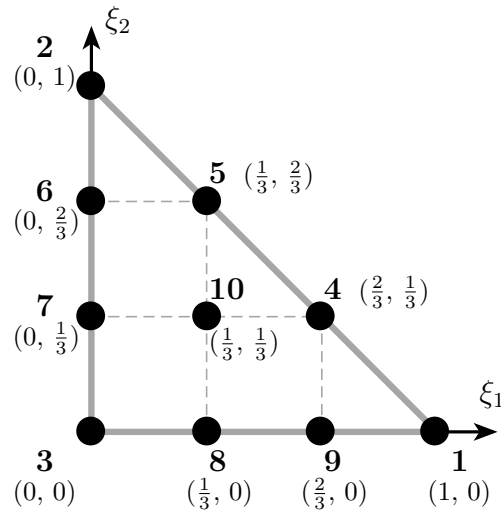


Figure 3.7: Linear ten-noded triangular finite element.

$$\phi_1(\boldsymbol{\xi}) = \frac{9}{2} \xi_1 \left(\xi_1 - \frac{1}{3} \right) \left(\xi_1 - \frac{2}{3} \right) \quad (3.32)$$

$$\phi_2(\boldsymbol{\xi}) = \frac{9}{2} \xi_2 \left(\xi_2 - \frac{1}{3} \right) \left(\xi_2 - \frac{2}{3} \right) \quad (3.33)$$

$$\phi_3(\boldsymbol{\xi}) = \frac{9}{2} (1 - \xi_1 - \xi_2) \left(\frac{1}{3} - \xi_1 - \xi_2 \right) \left(\frac{2}{3} - \xi_1 - \xi_2 \right) \quad (3.34)$$

$$\phi_4(\boldsymbol{\xi}) = \frac{27}{2} \xi_1 \xi_2 \left(\xi_1 - \frac{1}{3} \right) \quad (3.35)$$

$$\phi_5(\boldsymbol{\xi}) = \frac{27}{2} \xi_1 \xi_2 \left(\xi_2 - \frac{1}{3} \right) \quad (3.36)$$

$$\phi_6(\boldsymbol{\xi}) = \frac{27}{2} (1 - \xi_1 - \xi_2) \left(\xi_2 - \frac{1}{3} \right) \xi_2 \quad (3.37)$$

$$\phi_7(\boldsymbol{\xi}) = \frac{27}{2} (1 - \xi_1 - \xi_2) \left(\frac{2}{3} - \xi_1 - \xi_2 \right) \xi_2 \quad (3.38)$$

$$\phi_8(\boldsymbol{\xi}) = \frac{27}{2} (1 - \xi_1 - \xi_2) \left(\frac{2}{3} - \xi_1 - \xi_2 \right) \xi_1 \quad (3.39)$$

$$\phi_9(\boldsymbol{\xi}) = \frac{27}{2} (1 - \xi_1 - \xi_2) \left(\xi_1 - \frac{1}{3} \right) \xi_1 \quad (3.40)$$

$$\phi_{10}(\boldsymbol{\xi}) = 27 (1 - \xi_1 - \xi_2) \xi_1 \xi_2 \quad (3.41)$$

3.1.3 Quadrilaterals

LQ4

The linear four-noded quadrilateral finite element has shape functions with the general form $\phi(\boldsymbol{\xi}) = A + B\xi_1 + C\xi_2 + D\xi_1\xi_2$.

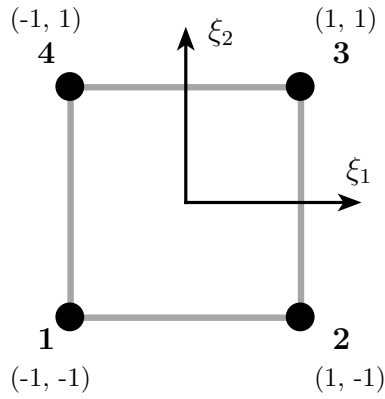


Figure 3.8: Linear four-noded quadrilateral finite element.

$$\phi_1(\boldsymbol{\xi}) = \frac{1}{4} (1 - \xi_1) (1 - \xi_2) \quad (3.42)$$

$$\phi_2(\boldsymbol{\xi}) = \frac{1}{4} (1 + \xi_1) (1 - \xi_2) \quad (3.43)$$

$$\phi_3(\boldsymbol{\xi}) = \frac{1}{4} (1 + \xi_1) (1 + \xi_2) \quad (3.44)$$

$$\phi_4(\boldsymbol{\xi}) = \frac{1}{4} (1 - \xi_1) (1 + \xi_2) \quad (3.45)$$

LQ8

The quadratic eight-noded quadrilateral finite element has shape functions with the general form $\phi(\boldsymbol{\xi}) = A + B\xi_1 + C\xi_2 + D\xi_1\xi_2 + E\xi_1^2 + F\xi_2^2 + G\xi_1^2\xi_2 + H\xi_1\xi_2^2$. This is often referred to as the “serendipity” quadrilateral element [explanation to come].

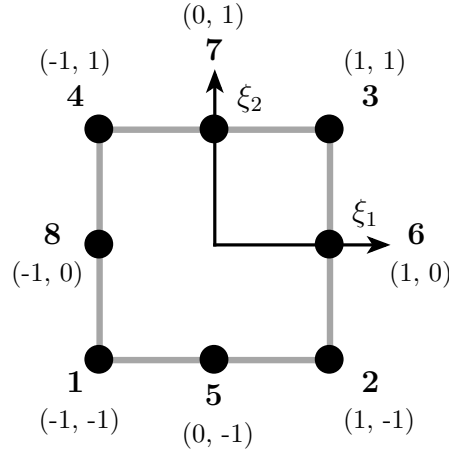


Figure 3.9: Linear eight-noded quadrilateral finite element.

$$\phi_1(\boldsymbol{\xi}) = \frac{1}{4} (1 - \xi_1) (1 - \xi_2) (-1 - \xi_1 - \xi_2) \quad (3.46)$$

$$\phi_2(\boldsymbol{\xi}) = \frac{1}{4} (1 + \xi_1) (1 - \xi_2) (-1 + \xi_1 - \xi_2) \quad (3.47)$$

$$\phi_3(\boldsymbol{\xi}) = \frac{1}{4} (1 + \xi_1) (1 + \xi_2) (-1 + \xi_1 + \xi_2) \quad (3.48)$$

$$\phi_4(\boldsymbol{\xi}) = \frac{1}{4} (1 - \xi_1) (1 + \xi_2) (-1 - \xi_1 + \xi_2) \quad (3.49)$$

$$\phi_5(\boldsymbol{\xi}) = \frac{1}{2} (1 - \xi_1^2) (1 - \xi_2) \quad (3.50)$$

$$\phi_6(\boldsymbol{\xi}) = \frac{1}{2} (1 + \xi_1) (1 - \xi_2^2) \quad (3.51)$$

$$\phi_7(\boldsymbol{\xi}) = \frac{1}{2} (1 - \xi_1^2) (1 + \xi_2) \quad (3.52)$$

$$\phi_8(\boldsymbol{\xi}) = \frac{1}{2} (1 - \xi_1) (1 - \xi_2^2) \quad (3.53)$$

3.2 Plasticity

3.2.1 Incremental Initial Value Problem for Elastoplasticity with Isotropic Hardening

To motivate the time stepping components of the implementation algorithm, consider smooth function $f(x) : \mathbb{R} \rightarrow \mathbb{R}$. With initial condition $x(0) = x_0$ and the notation $x_n \approx x(t_n)$, f can be approximated as

$$\dot{x}(t) = f(x(t)), \quad (3.54)$$

$$\approx \frac{x_{n+1} - x_n}{\Delta t} = f(x_{n+\theta}), \quad \text{where } \theta \in [0, 1], \quad \text{and} \quad (3.55)$$

$$x_{n+\theta} = (1 - \theta)x_n + \theta x_{n+1}. \quad (3.56)$$

The value θ controls the stability and accuracy of the time stepping algorithm.

$$\begin{aligned} \theta = 0 & \quad \text{forward (explicit) Euler,} \\ \theta = \frac{1}{2} & \quad \text{midpoint rule,} \\ \theta = 1 & \quad \text{backward (implicit) Euler.} \end{aligned} \quad (3.57)$$

With these notions at hand and use of the implicit Euler, we use (3.55) for evolution in the plastic flow rule.

$$\dot{\varepsilon}^p \approx \frac{\varepsilon_{n+1}^p - \varepsilon_n^p}{\Delta t} = [\dot{\gamma} \text{sign}(\sigma)]_{n+\theta}, \quad (3.58)$$

(now with $\theta = 1$)

$$= \dot{\gamma}_{n+1} \text{sign}(\sigma_{n+1}). \quad (3.59)$$

Using (3.55) again for $\dot{\gamma}$ gives

$$\dot{\gamma} \approx \frac{\gamma_{n+1} - \gamma_n}{\Delta t} = \dot{\gamma}_{n+\theta}, \quad (3.60)$$

(now with $\theta = 1$)

$$= \dot{\gamma}_{n+1}. \quad (3.61)$$

Thus

$$\dot{\varepsilon}^p \approx \frac{\varepsilon_{n+1}^p - \varepsilon_n^p}{\Delta t} = \frac{\gamma_{n+1} - \gamma_n}{\Delta t} \text{sign}(\sigma_{n+1}). \quad (3.62)$$

Adopting the notation $\gamma_{n+1} - \gamma_n = \Delta\gamma$,

$$\boxed{\varepsilon_{n+1}^p = \varepsilon_n^p + \Delta\gamma \text{sign}(\sigma_{n+1})}. \quad (3.63)$$

Analogous arguments with the hardening variable evolution $\dot{\alpha} = \dot{\gamma}$ lead to

$$\boxed{\alpha_{n+1} = \alpha_n + \Delta\gamma}. \quad (3.64)$$

3.2.2 Trial Elastic State and Plastic Return-Mapping

Consider now a trial state that is purely elastic, with all plastic flow “frozen” in time. The incremental displacement $\Delta \mathbf{u}$ creates an incremental strain $\Delta \varepsilon \equiv \varepsilon_{n+1} - \varepsilon_n$. Then the trial state functions are denoted as

$$\sigma_{n+1}^{\text{tr}} = E (\varepsilon_{n+1} - \varepsilon_n^{\text{p}}) \equiv \sigma_n + E \Delta \varepsilon \quad (3.65)$$

$$\Phi_{n+1}^{\text{tr}} = |\sigma_{n+1}^{\text{tr}}| - (\sigma_y + H \alpha_n) \quad (3.66)$$

$$\varepsilon_{n+1}^{\text{p tr}} = \varepsilon_n^{\text{p}} \quad (3.67)$$

$$\alpha_{n+1}^{\text{tr}} = \alpha_n \quad (3.68)$$

With the known initial conditions $\{\varepsilon_n^{\text{p}}, \alpha_n\}$, the strain is incremented $\Delta \varepsilon$ and the resulting trial yield function is tested for the occurrence of a strictly elastic step or an elasto-plastic step. Specifically,

$$\begin{aligned} &\text{if } \Phi_{n+1}^{\text{tr}} \leq 0, \text{ then } \Delta \gamma = 0 \text{ and} \\ &\left. \begin{aligned} \sigma_{n+1} &= \sigma_{n+1}^{\text{tr}} \\ \Phi_{n+1} &= \Phi_{n+1}^{\text{tr}} \\ \varepsilon_{n+1}^{\text{p}} &= \varepsilon_{n+1}^{\text{p tr}} \\ \alpha_{n+1} &= \alpha_{n+1}^{\text{tr}} \end{aligned} \right\} \text{ (elastic step).} \end{aligned} \quad (3.69)$$

Otherwise, the return-mapping algorithm is used for the elasto-plastic step,

$$\begin{aligned} &\text{if } \Phi_{n+1}^{\text{tr}} > 0, \text{ then } \Delta \gamma > 0 \text{ and} \\ &\left. \begin{aligned} \sigma_{n+1} &= \sigma_{n+1}^{\text{tr}} - E \Delta \gamma \text{ sign}(\sigma_{n+1}^{\text{tr}}) \\ \Phi_{n+1} &= 0 \\ \varepsilon_{n+1}^{\text{p}} &= \varepsilon_n^{\text{p}} + \Delta \gamma \text{ sign}(\sigma_{n+1}^{\text{tr}}) \\ \alpha_{n+1} &= \alpha_n + \Delta \gamma \end{aligned} \right\} \text{ (ep step).} \end{aligned} \quad (3.70)$$

The results in (3.70) are derived as follows. First, find the following two useful relationships between σ_{n+1}^{tr} and σ_{n+1} .

$$\sigma_{n+1} = E (\varepsilon_{n+1} - \varepsilon_{n+1}^{\text{p}}) \quad (3.71)$$

$$= E (\varepsilon_{n+1} - \varepsilon_n^{\text{p}} + \varepsilon_n^{\text{p}} - \varepsilon_{n+1}^{\text{p}}), \quad (3.72)$$

$$\stackrel{(3.63, 3.65)}{=} \sigma_{n+1}^{\text{tr}} - E \Delta \gamma \text{ sign}(\sigma_{n+1}^{\text{tr}}). \quad (3.73)$$

Thus

$$\text{sign}(\sigma_{n+1}) (|\sigma_{n+1}| + E \Delta \gamma) = \text{sign}(\sigma_{n+1}^{\text{tr}}) |\sigma_{n+1}^{\text{tr}}|. \quad (3.74)$$

Since $E > 0$ and $\Delta\gamma > 0$, we require

$$\boxed{\text{sign}(\sigma_{n+1}) = \text{sign}(\sigma_{n+1}^{\text{tr}})}, \quad (3.75)$$

and

$$\boxed{|\sigma_{n+1}| + E\Delta\gamma = |\sigma_{n+1}^{\text{tr}}|}. \quad (3.76)$$

Next, determine the incremental plastic multiplier $\Delta\gamma$ using the discrete versions of the yield function

$$\Phi_{n+1} \stackrel{(2.83)}{=} |\sigma_{n+1}| - (\sigma_y + H\alpha_{n+1}) \stackrel{\text{set}}{=} 0, \quad (3.77)$$

$$\begin{aligned} &\stackrel{(3.76)}{=} |\sigma_{n+1}^{\text{tr}}| - E\Delta\gamma - (\sigma_y + H\alpha_n) \\ &\quad - H(\alpha_{n+1} - \alpha_n), \end{aligned} \quad (3.78)$$

$$\stackrel{(3.66, 3.64)}{=} \Phi_{n+1}^{\text{tr}} - E\Delta\gamma - H\Delta\gamma, \quad (3.79)$$

gives

$$\boxed{\Delta\gamma = \frac{\Phi_{n+1}^{\text{tr}}}{E + H}}. \quad (3.80)$$

Then all variables Eq. (3.70) are known. Equation (3.70)₁, rewritten as,

$$\sigma_{n+1} = \left[1 - \frac{E\Delta\gamma}{|\sigma_{n+1}^{\text{tr}}|} \right] \sigma_{n+1}^{\text{tr}}, \quad (3.81)$$

shows that the final stress state σ_{n+1} may be interpreted as the **projection** of the trial stress state σ_{n+1}^{tr} on the yield function. Figure 3.10 (b) illustrates this concept.

3.2.3 Linearization

The Taylor series expansion of real function $f(x)$ about a point $x = a$ is applied to the residual function $\mathbf{r}(\mathbf{u})$ as follows,

$$\begin{aligned} f(x) &= f(a) \\ &\quad + f'(a)(x - a) \\ &\quad + \frac{f''(a)}{2!}(x - a)^2 \\ &\quad + \dots \\ &\quad + \frac{f^{(n)}(a)}{n!}(x - a)^n \end{aligned} \quad (3.82)$$

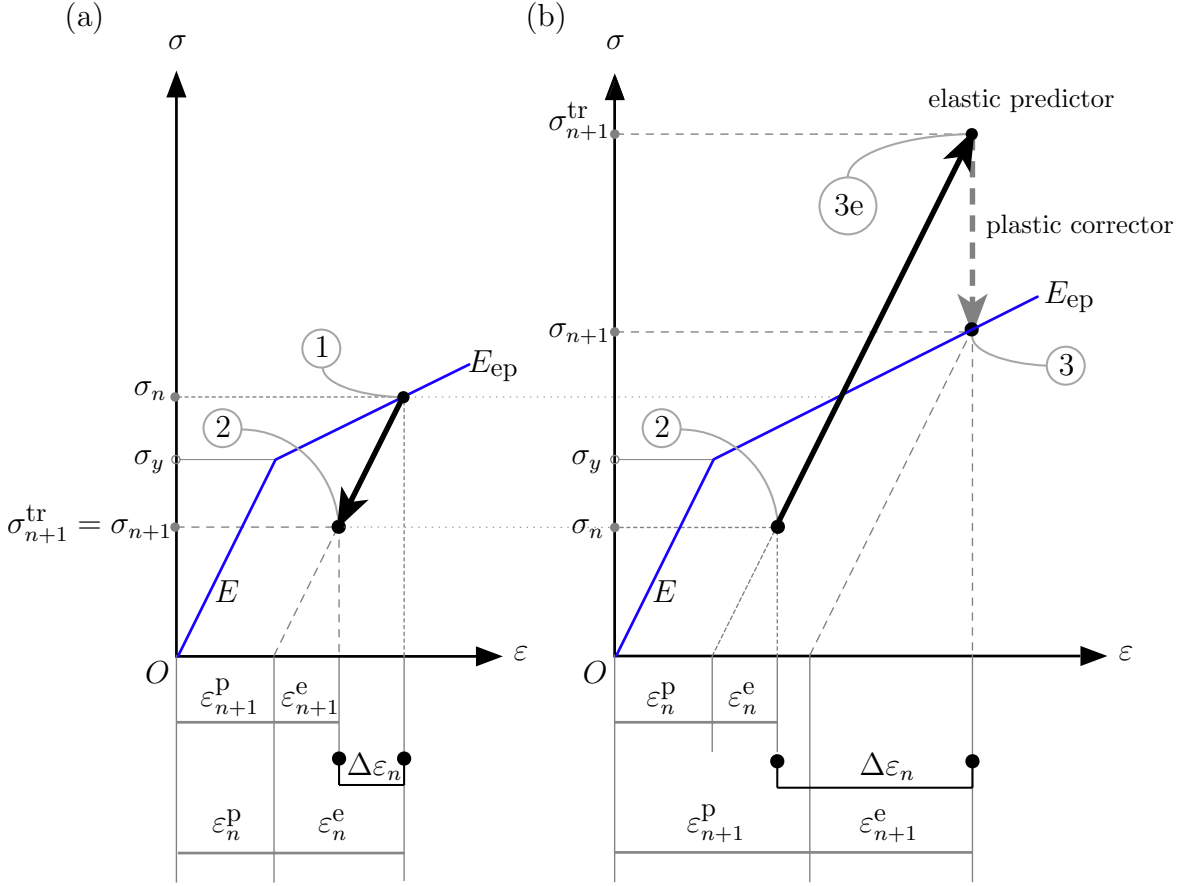


Figure 3.10: (a) Unloading step from state 1 to state 2 and (b) loading step from state 2, to state 3e (elastic prediction), and to state 3.

Thus, with combining all higher order terms, abbreviated as (H.O.T.),

$$\mathbf{r}(\mathbf{u} + \Delta\mathbf{u}) = \mathbf{r}(\mathbf{u}) + \frac{\partial \mathbf{r}(\mathbf{u})}{\partial \mathbf{u}} \Delta\mathbf{u} + \text{H.O.T.}, \quad (3.83)$$

and

$$\mathbf{r}^{(i+1)} = \mathbf{r}^{(i)} + \frac{\partial \mathbf{r}^{(i)}}{\partial \mathbf{u}^{(i)}} \Delta\mathbf{u}^{(i)} + \text{H.O.T.} \quad (3.84)$$

This iterative approach is the groundwork for the Newton-Raphson iterative procedure. At the initial iteration, $i = 0$, the initial guess is set equal to the known quantity at the previously converged time step, $\mathbf{u}_{n+1}^{(0)} = \mathbf{u}_n$. Then, linearization about the current iterative step is taken, and updates are determined, as explicated in Box 1. Figure 3.11 shows a graphical representation of this procedure.

$\mathbf{u}_{n+1}^{(0)} = \mathbf{u}_n$ initial guess as converged solution at t_n

$$\left[\frac{-\partial \mathbf{r}(\mathbf{u}_{n+1}^{(0)})}{\partial \mathbf{u}_{n+1}^{(0)}} \right]^{-1} \mathbf{r}(\mathbf{u}_{n+1}^{(0)}) = \Delta \mathbf{u}_{n+1}^{(0)}$$

$$\implies \Delta \mathbf{u}_{n+1}^{(0)} + \mathbf{u}_{n+1}^{(0)} = \mathbf{u}_{n+1}^{(1)} \implies \frac{|\mathbf{r}(\mathbf{u}_{n+1}^{(1)})|}{|\mathbf{f}_{n+1}^{\text{ext}}|} \stackrel{?}{\leq} \text{TOL}$$

$$\left[\frac{-\partial \mathbf{r}(\mathbf{u}_{n+1}^{(1)})}{\partial \mathbf{u}_{n+1}^{(1)}} \right]^{-1} \mathbf{r}(\mathbf{u}_{n+1}^{(1)}) = \Delta \mathbf{u}_{n+1}^{(1)}$$

$$\implies \Delta \mathbf{u}_{n+1}^{(1)} + \mathbf{u}_{n+1}^{(1)} = \mathbf{u}_{n+1}^{(2)} \implies \frac{|\mathbf{r}(\mathbf{u}_{n+1}^{(2)})|}{|\mathbf{f}_{n+1}^{\text{ext}}|} \stackrel{?}{\leq} \text{TOL}$$

$\vdots$ $\vdots$ $\vdots$

$$\dots \dots \implies \frac{|\mathbf{r}(\mathbf{u}_{n+1}^{(*)})|}{|\mathbf{f}_{n+1}^{\text{ext}}|} \leq \text{TOL}$$

$\mathbf{u}_{n+1} = \mathbf{u}_{n+1}^{(*)}$ update with converged solution

Box 1: Newton-Raphson iterative procedure, explicated for the first and second iterations.

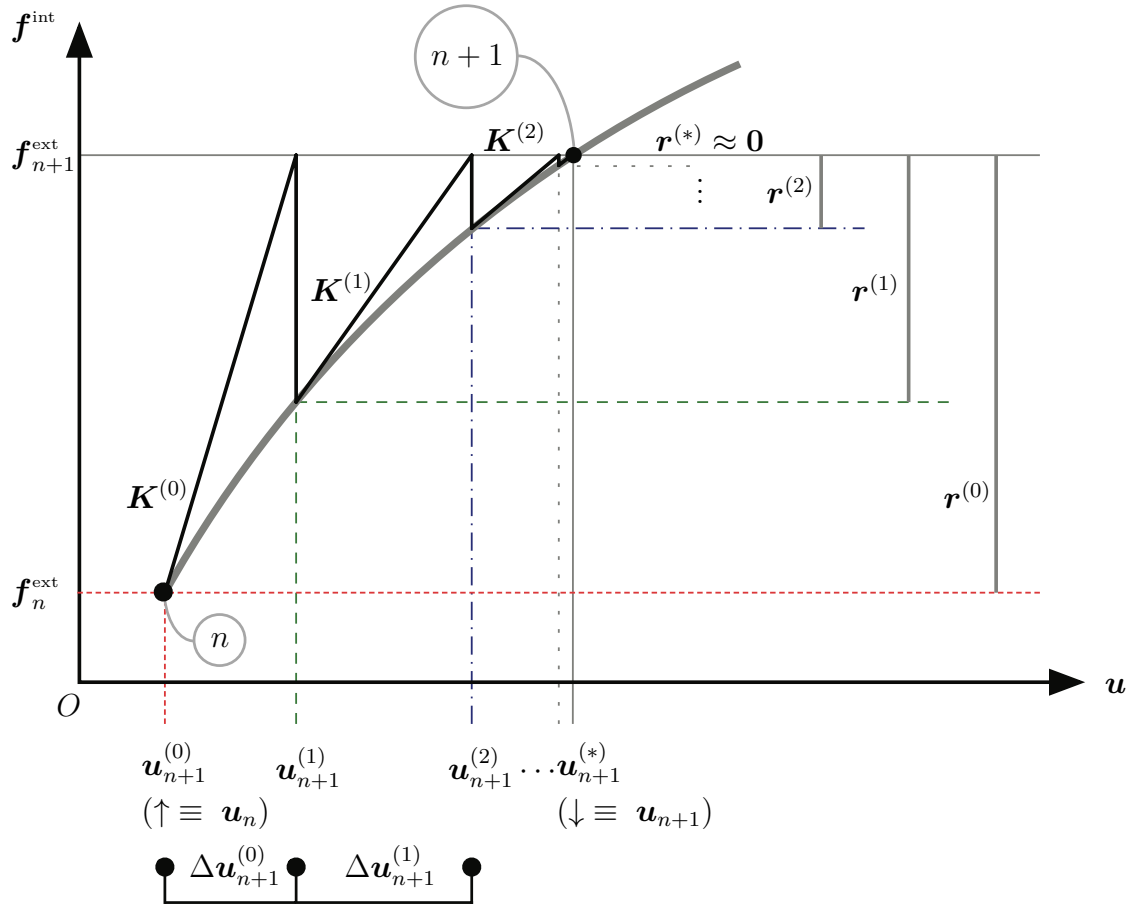


Figure 3.11: Newton-Raphson iterative procedure, shown graphically: The first iteration $K^{(0)}$ approximates the curve as tangent line at n . Successive tangents are constructed $K^{(i)}$ until the root at $n+1$ is found.

Time step loop ($n = 0$, WHILE $n \leq \text{NSTEPS}$)1. $i = 0$, CONVERGED = FALSE2. $\mathbf{u}_{n+1}^{(i)} = \mathbf{u}_n$ (prediction)3. **Iteration loop** (WHILE $i \leq \text{IMAX}$ AND CONVERGED IS FALSE)

- if $\frac{\left| \mathbf{r} \left(\mathbf{u}_{n+1}^{(i)} \right) \right|}{\left| \mathbf{f}_{n+1}^{\text{ext}} \right|} \leq \text{TOL} \quad \left| \mathbf{f}_{n+1}^{\text{ext}} \right| \neq 0$

- then

- (a) CONVERGED = TRUE

- (b) $\mathbf{u}_{n+1}^{(*)} = \mathbf{u}_{n+1}^{(i)}$

- (c) $\varepsilon_{n+1}^{\text{P} (*)} = \varepsilon_{n+1}^{\text{P} (i)}$

- (d) $\alpha_{n+1}^{(*)} = \alpha_{n+1}^{(i)}$

- else

- (a) $\left[\frac{-\partial \mathbf{r} \left(\mathbf{u}_{n+1}^{(i)} \right)}{\partial \mathbf{u}_{n+1}^{(i)}} \right]^{-1} \mathbf{r} \left(\mathbf{u}_{n+1}^{(i)} \right) = \Delta \mathbf{u}_{n+1}^{(i)}$ (form tangent and solve)

- (b) $\mathbf{u}_{n+1}^{(i+1)} = \mathbf{u}_{n+1}^{(i)} + \Delta \mathbf{u}_{n+1}^{(i)}$ (iteration update)

- (c) $i \leftarrow i + 1$

4. $\mathbf{u}_{n+1} = \mathbf{u}_{n+1}^{(*)}$ (time step update)5. $\varepsilon_{n+1}^{\text{P}} = \varepsilon_{n+1}^{\text{P} (*)}$ 6. $\alpha_{n+1} = \alpha_{n+1}^{(*)}$ 7. $n \leftarrow n + 1$

Box 2: Predictor/corrector time stepping algorithm applied to elasto-plastic equations of motion.